
Course Outline

1. Introduction
 2. Neuronal Signaling
 3. Visual System
 4. Neural Information Processing
 5. Decision making
 6. Multi-modal Sensory Systems
 7. Hippocampus, Spatial Cognition & Memory
 8. Value-based Decision Making
 9. Motor Systems
 10. Learning & Plasticity
 11. Attention and More
 12. Project Presentation
 13. Final Q&A
-

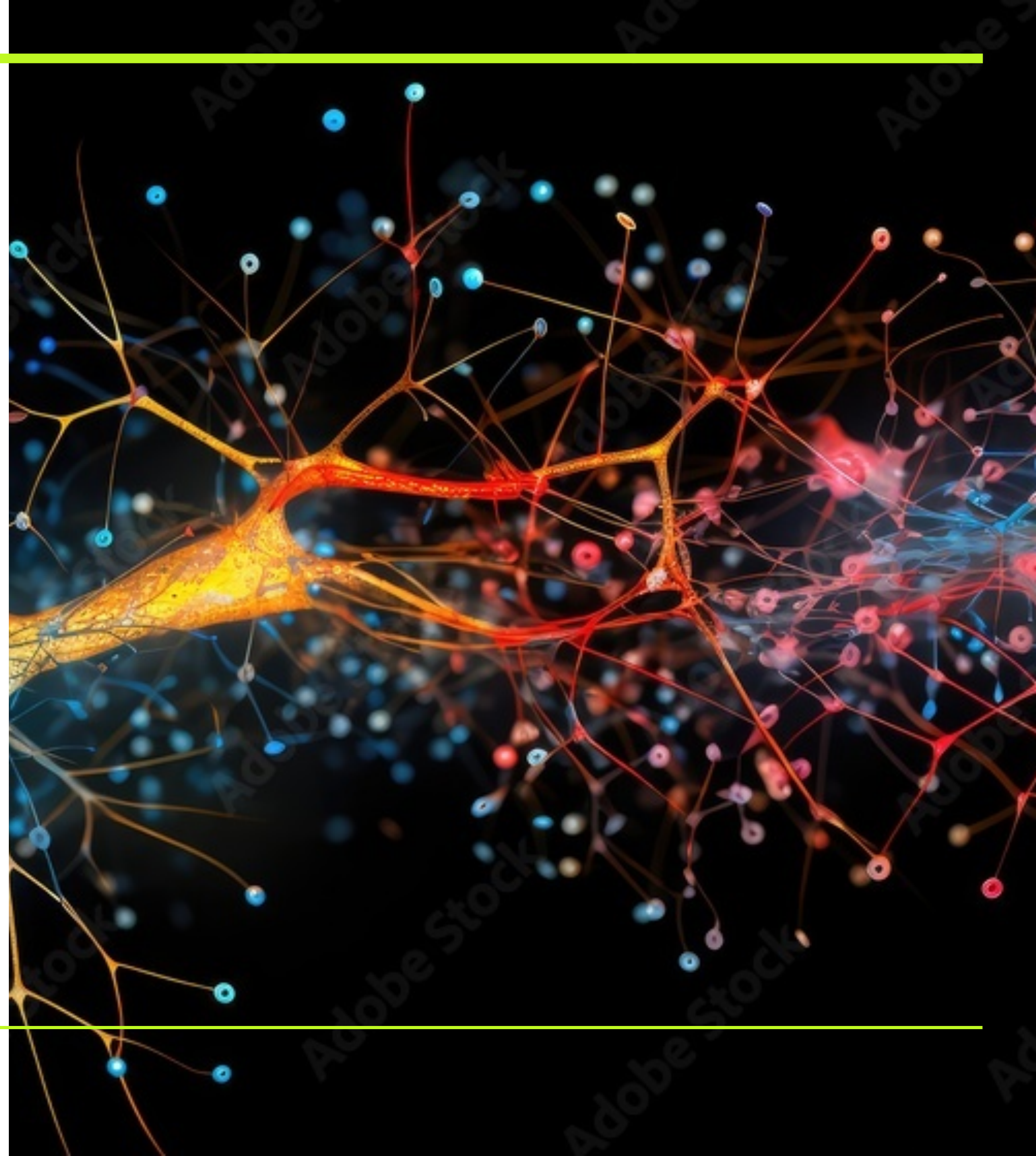
系统和计算神经科学导论

十、学习和可塑性

Learning & Plasticity

中国科学院脑科学与智能技术卓越创新中心（神经所） 杨天明 TYANG@ION.AC.CN

-
- ❖ **Overview**
 - ❖ **Synaptic Mechanism**
 - ❖ **Hebbian Learning**
 - ❖ **Hopfield Network**

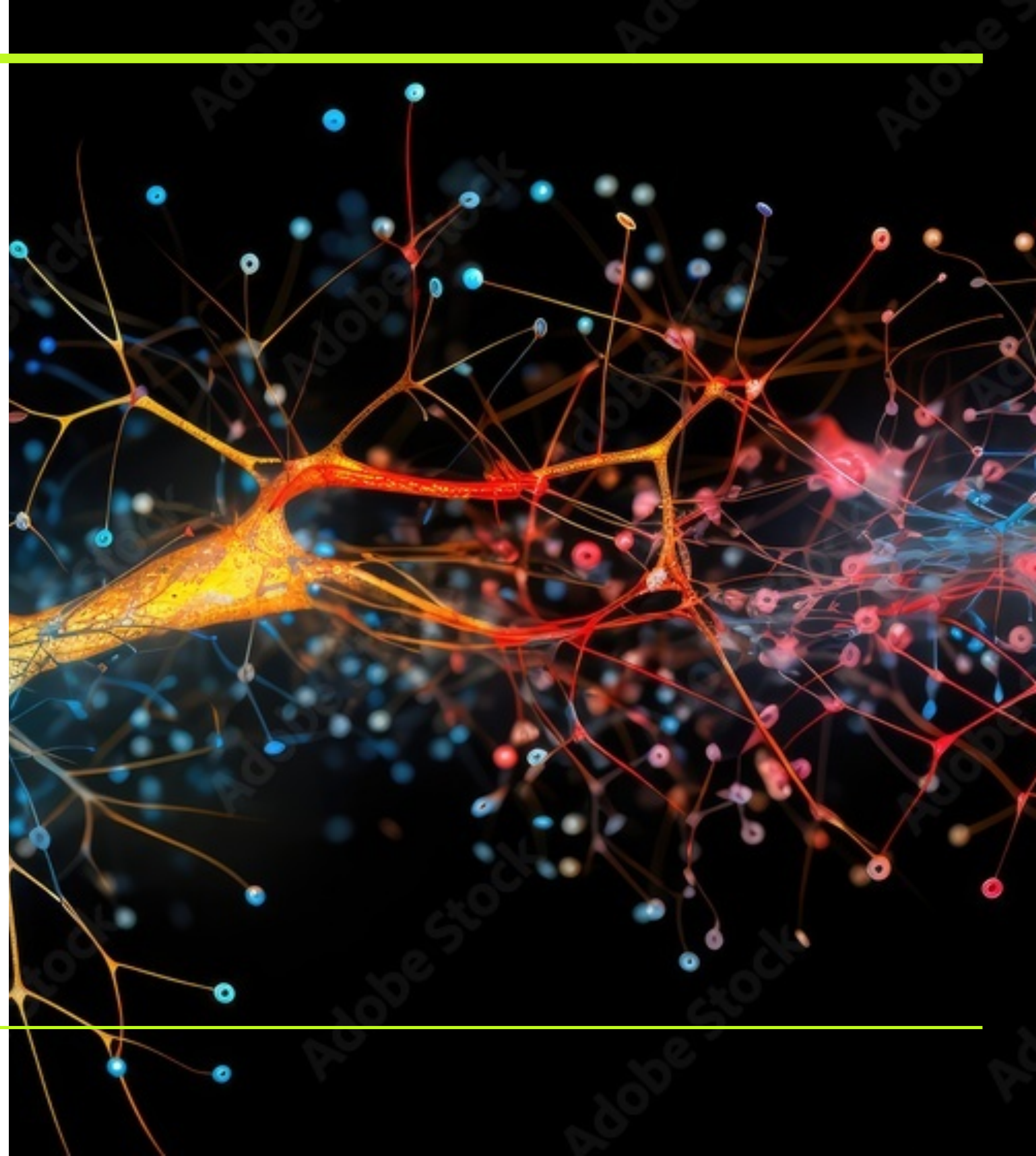


❖ Overview

❖ Synaptic Mechanism

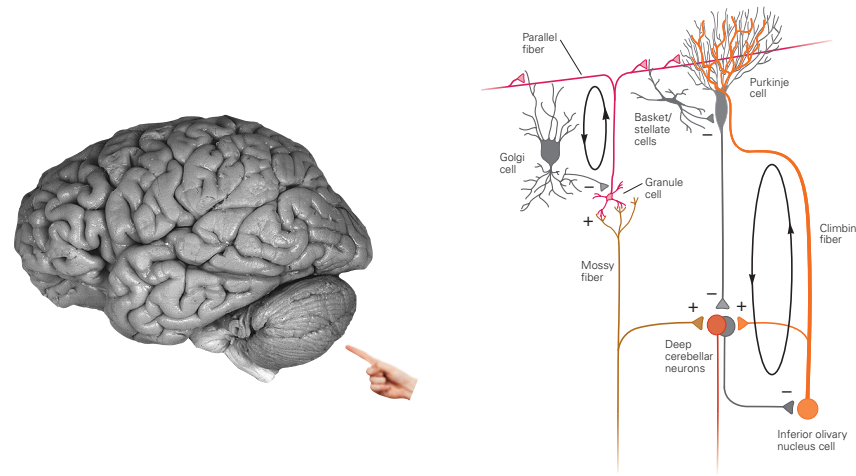
❖ Hebbian Learning

❖ Hopfield Network



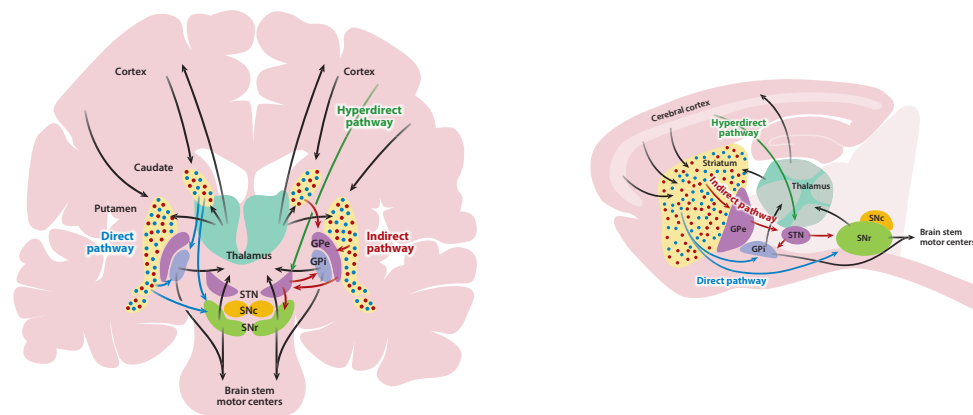
Learning in the brain

- Supervised



Cerebellum

- Reinforcement Learning



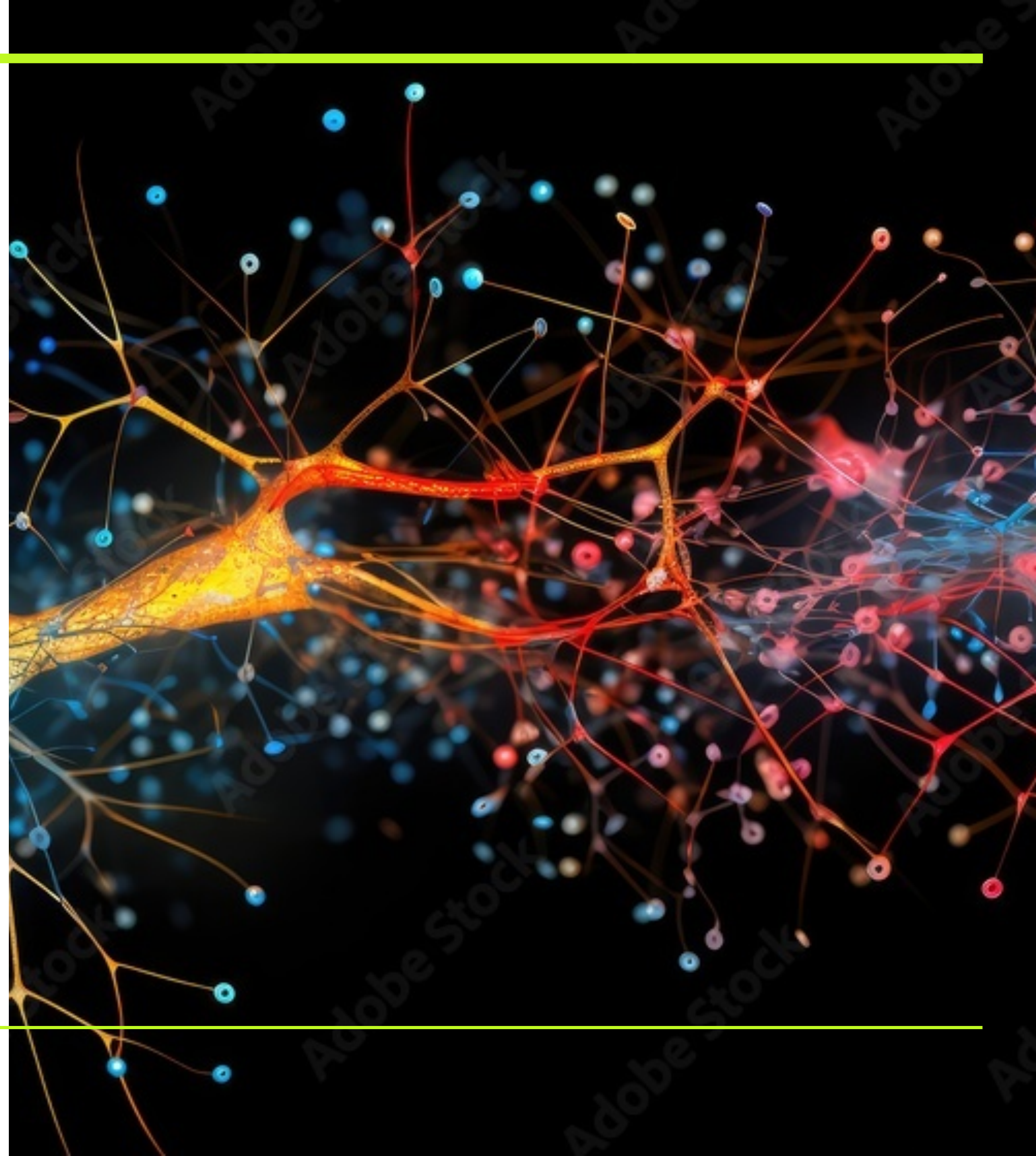
Basal ganglia

- Unsupervised

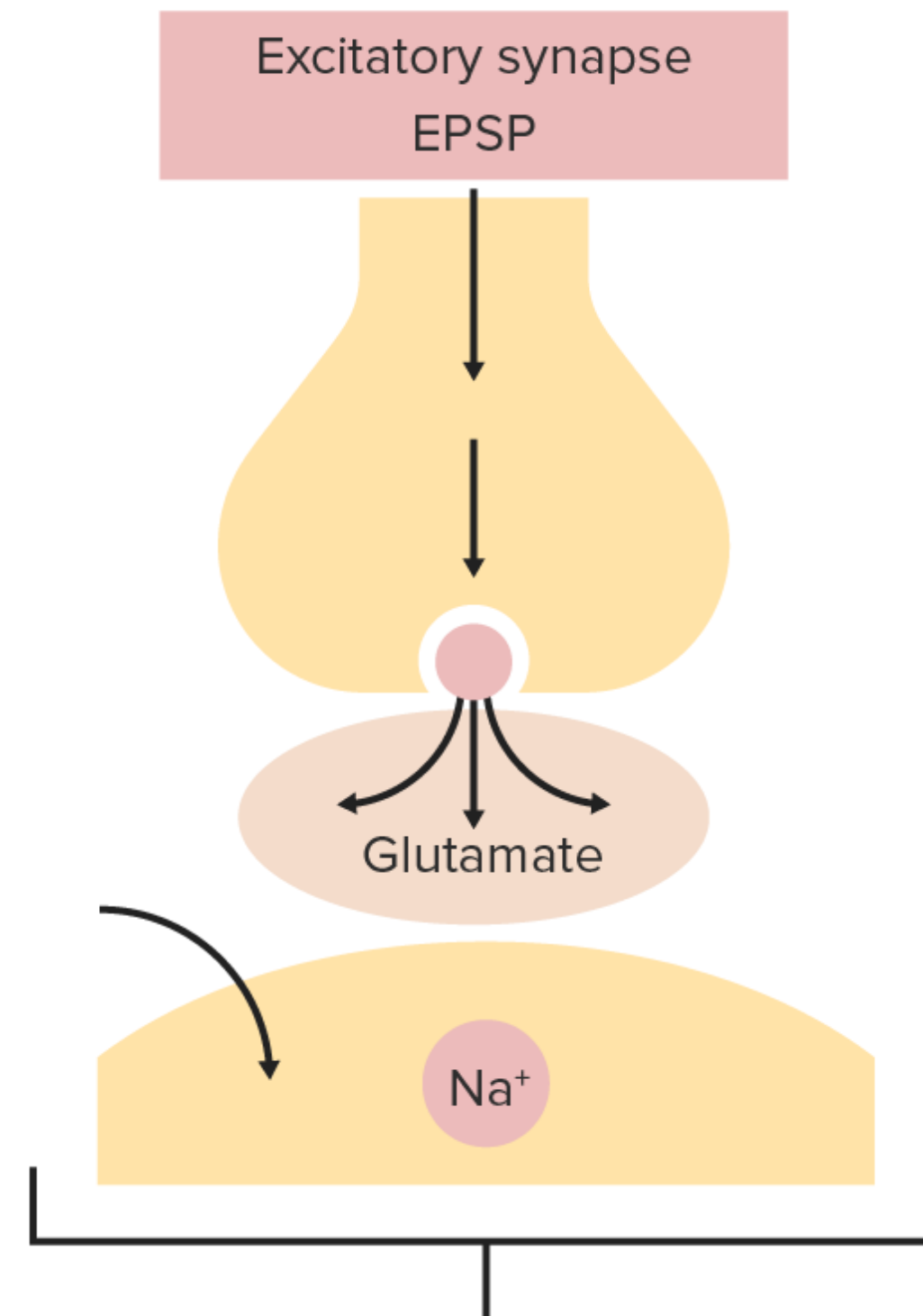


All over the brain

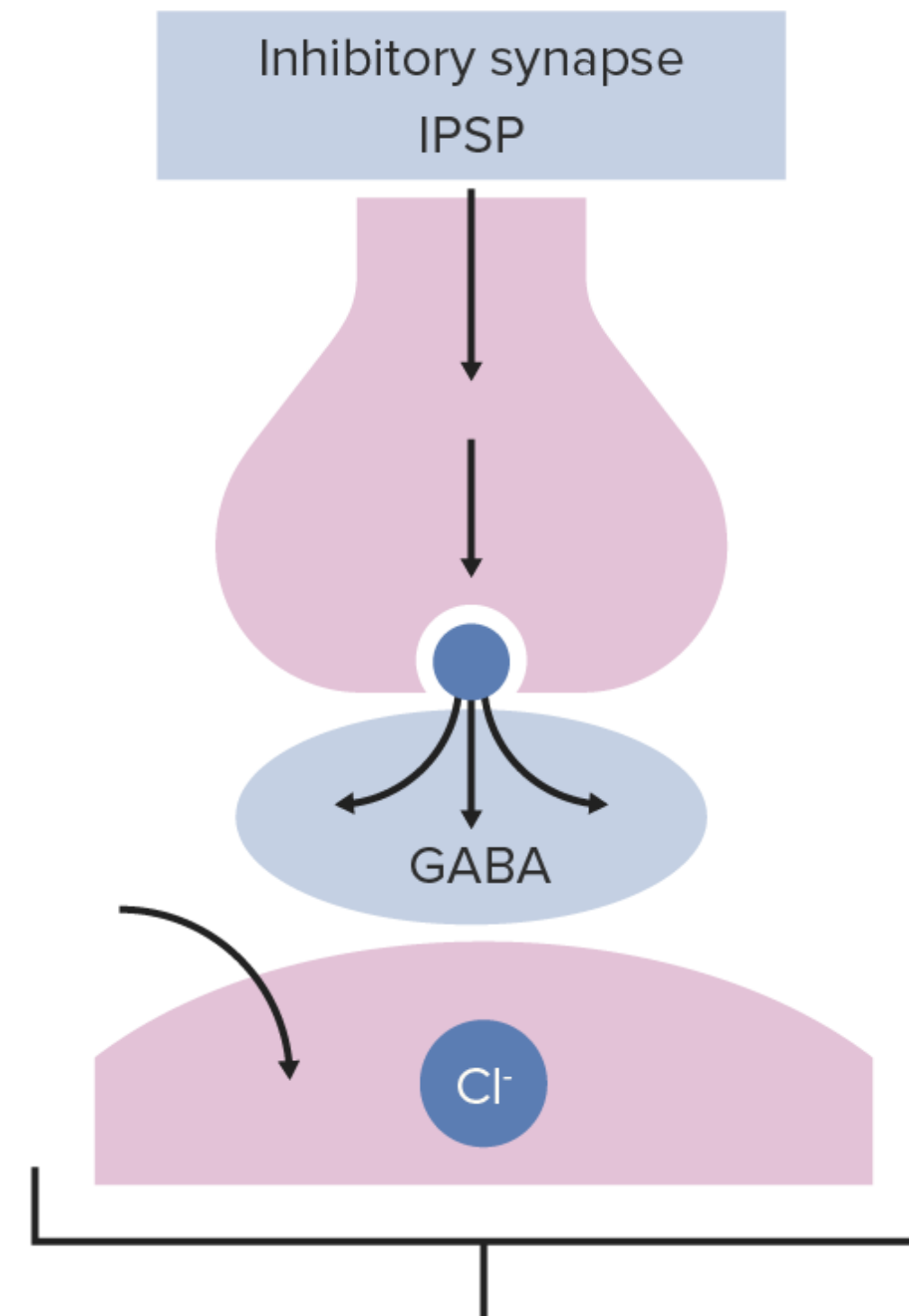
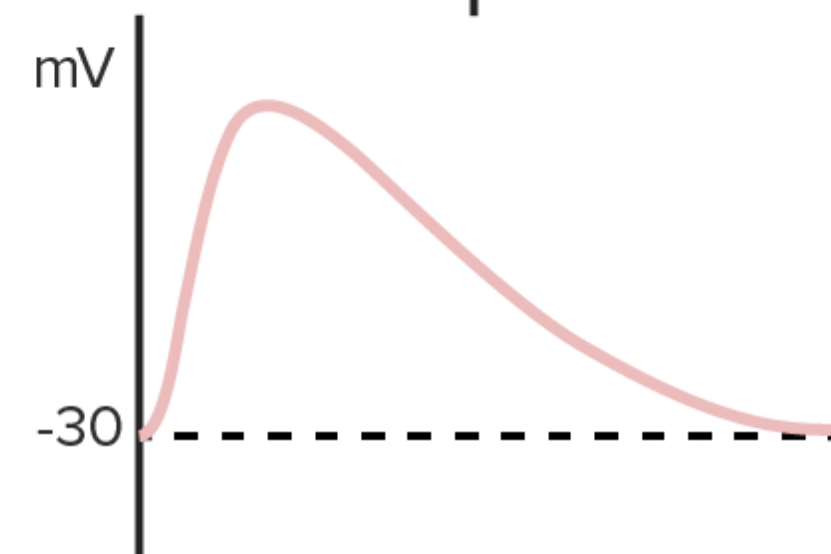
-
- ❖ Overview
 - ❖ **Synaptic Mechanism**
 - ❖ Hebbian Learning
 - ❖ Hopfield Network
-



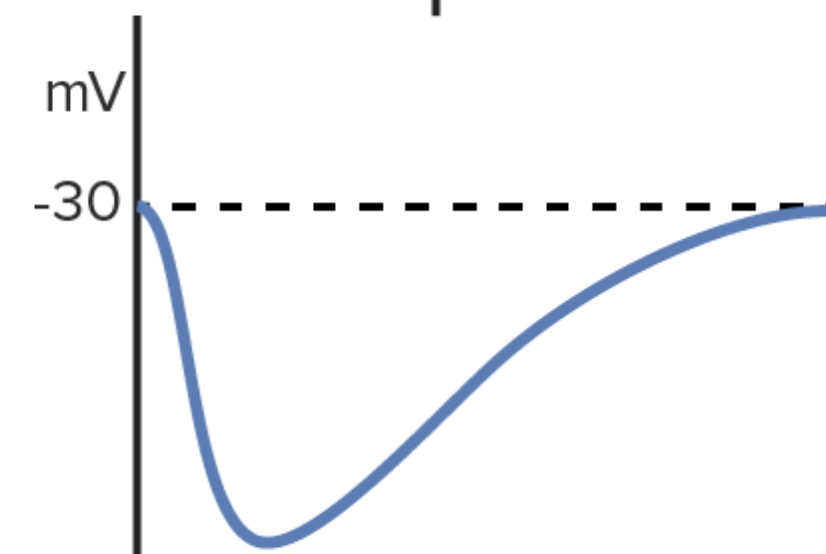
EPSP & IPSP



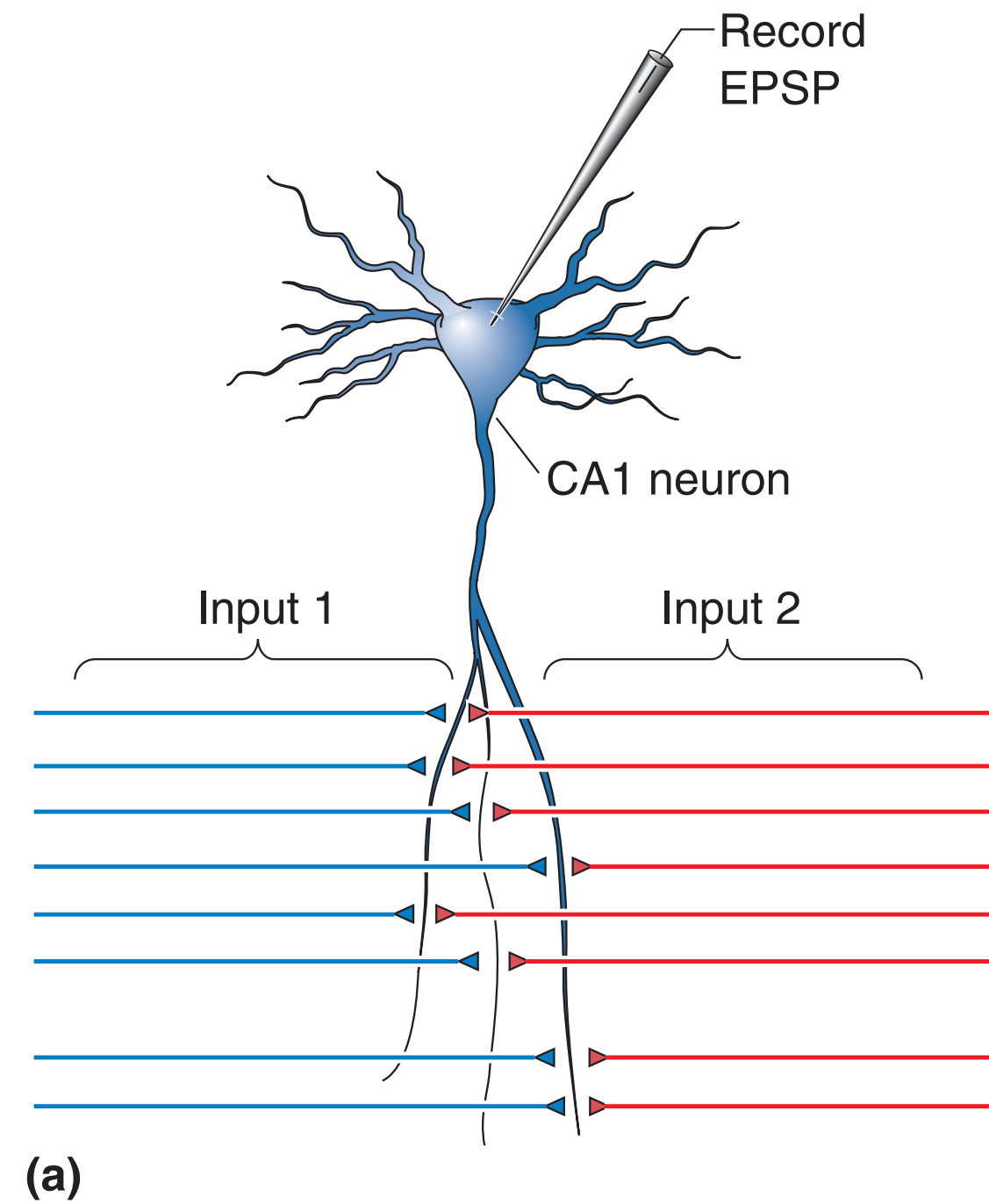
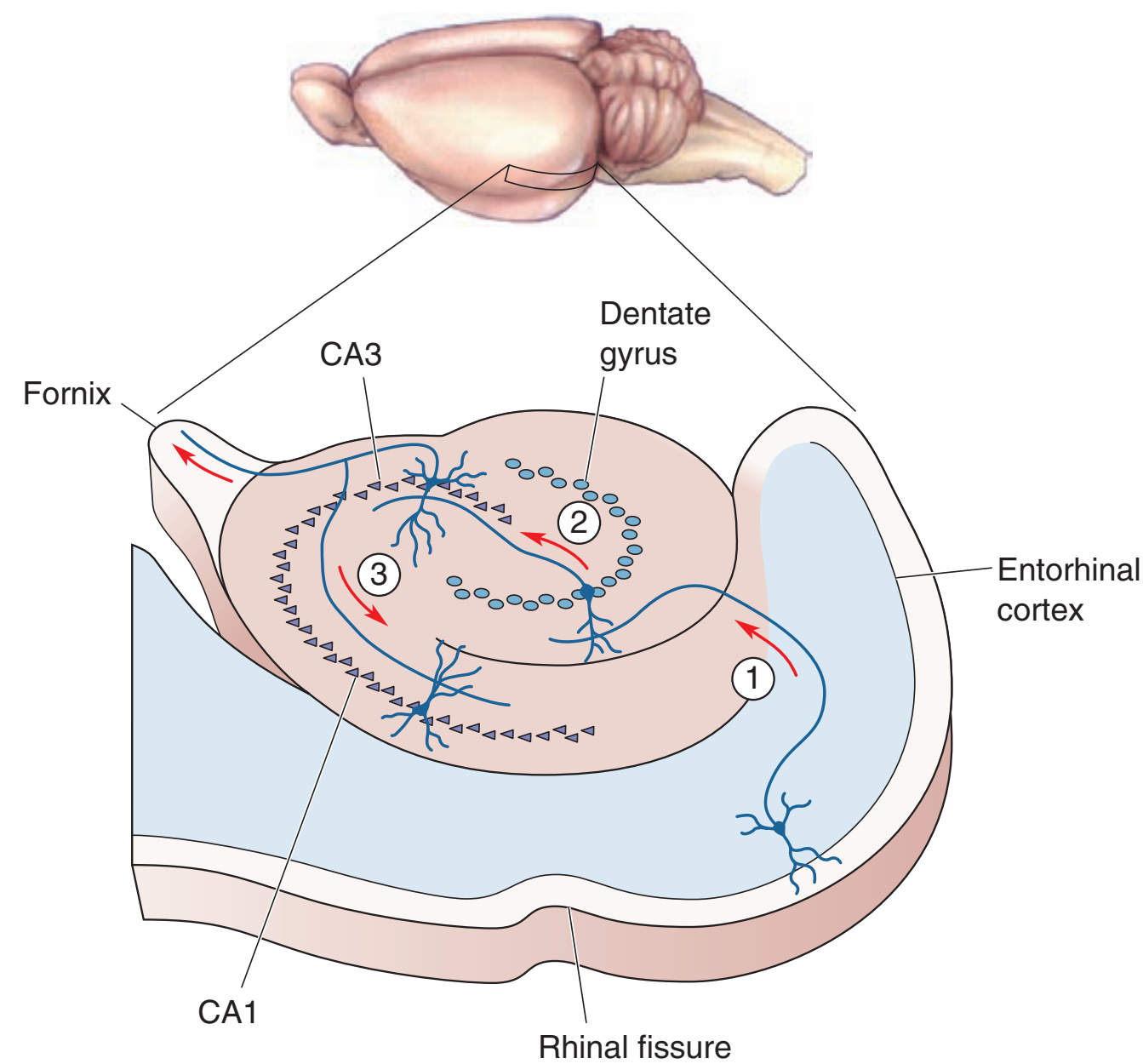
Excitatory
Post-Synapt
Potential



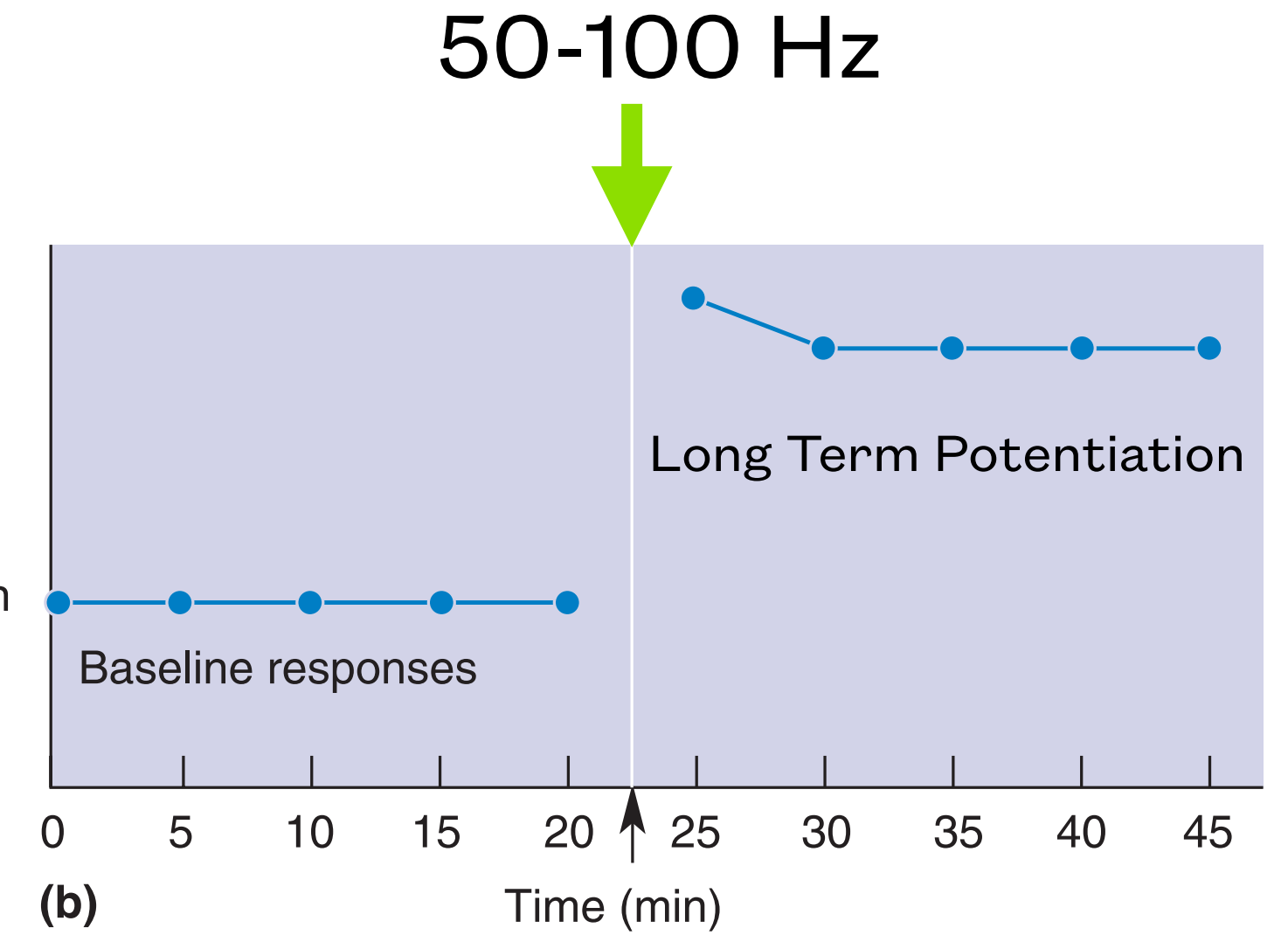
Inhibitory
Post-Synaptic
Potential



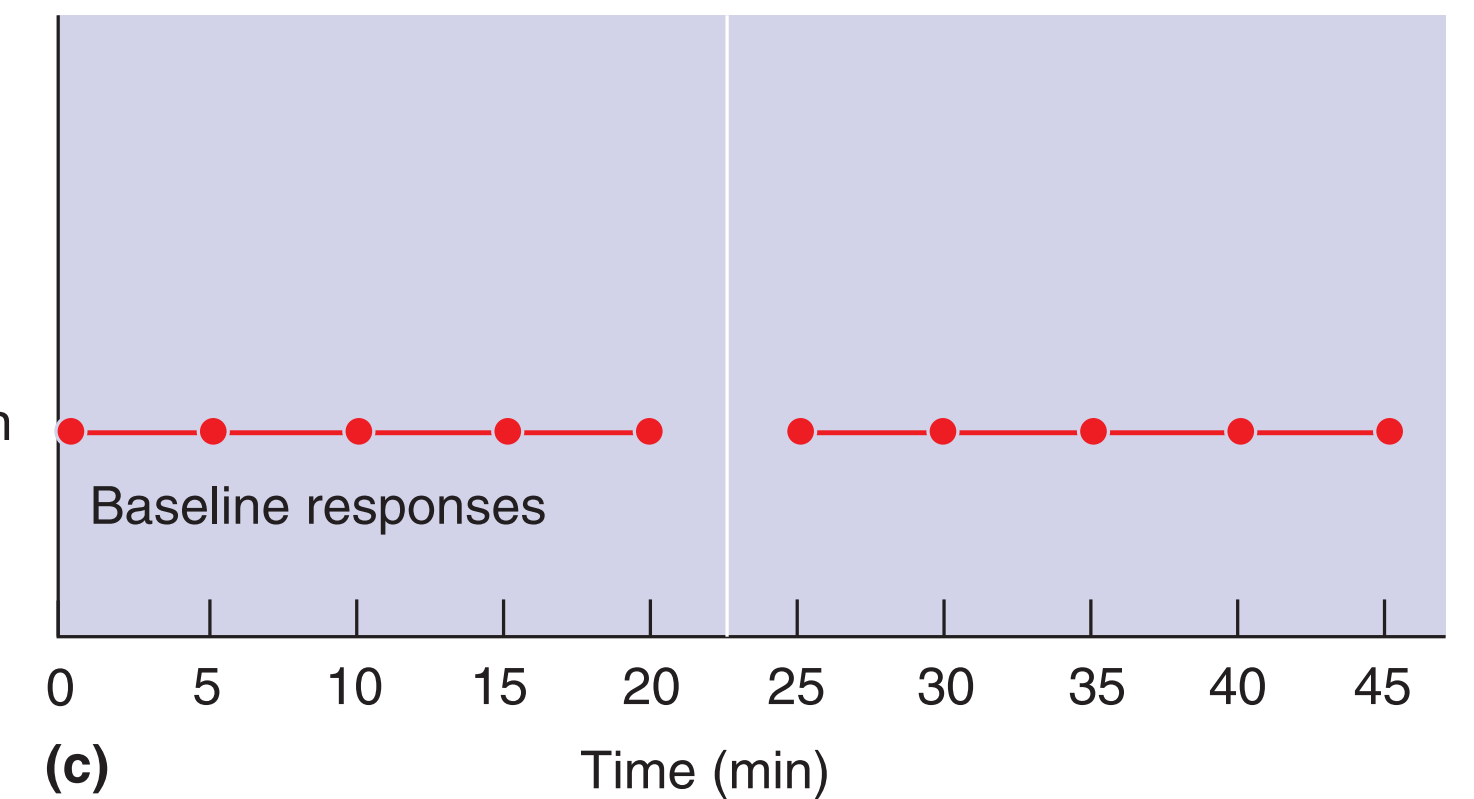
LTP in Hippocampus



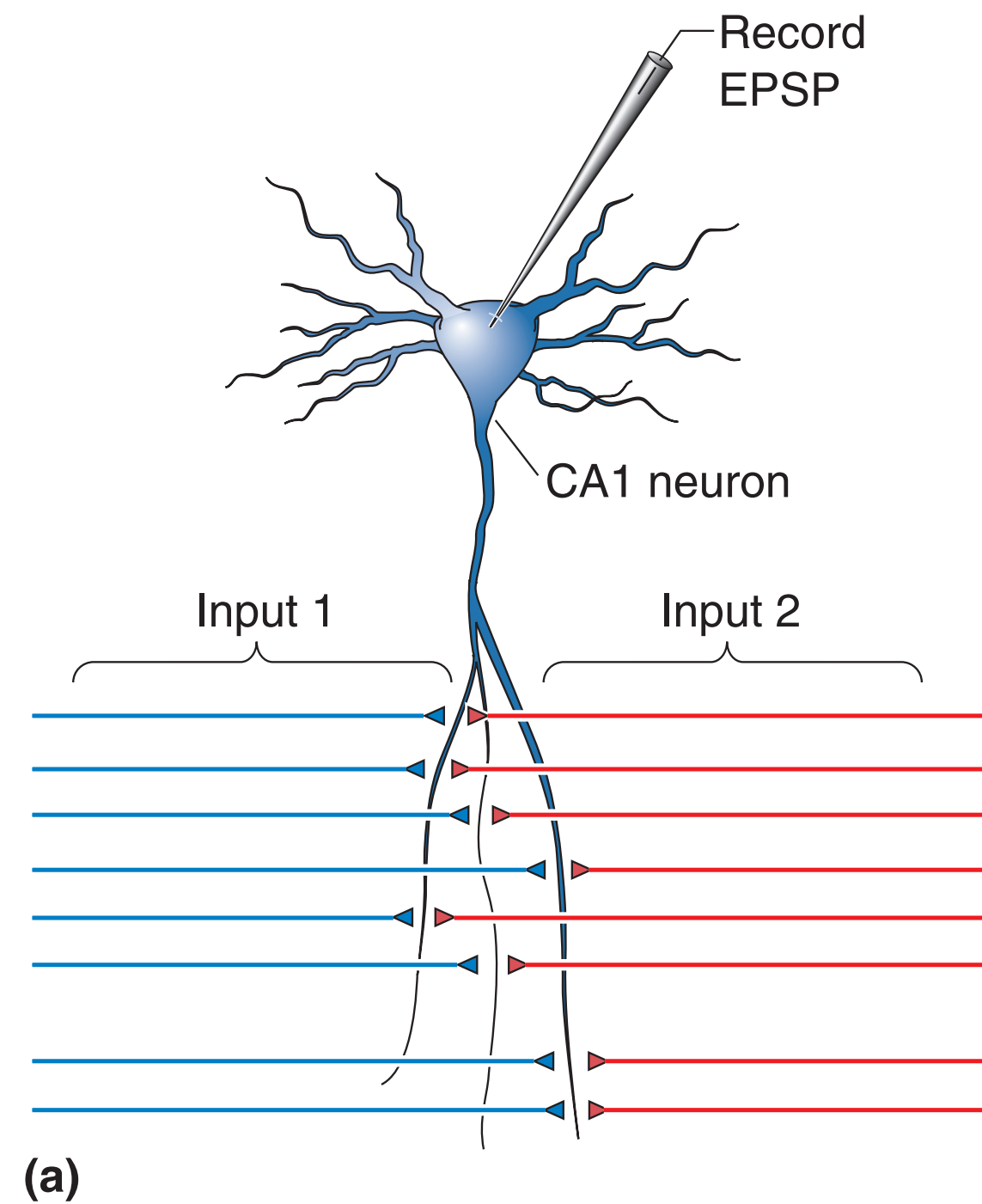
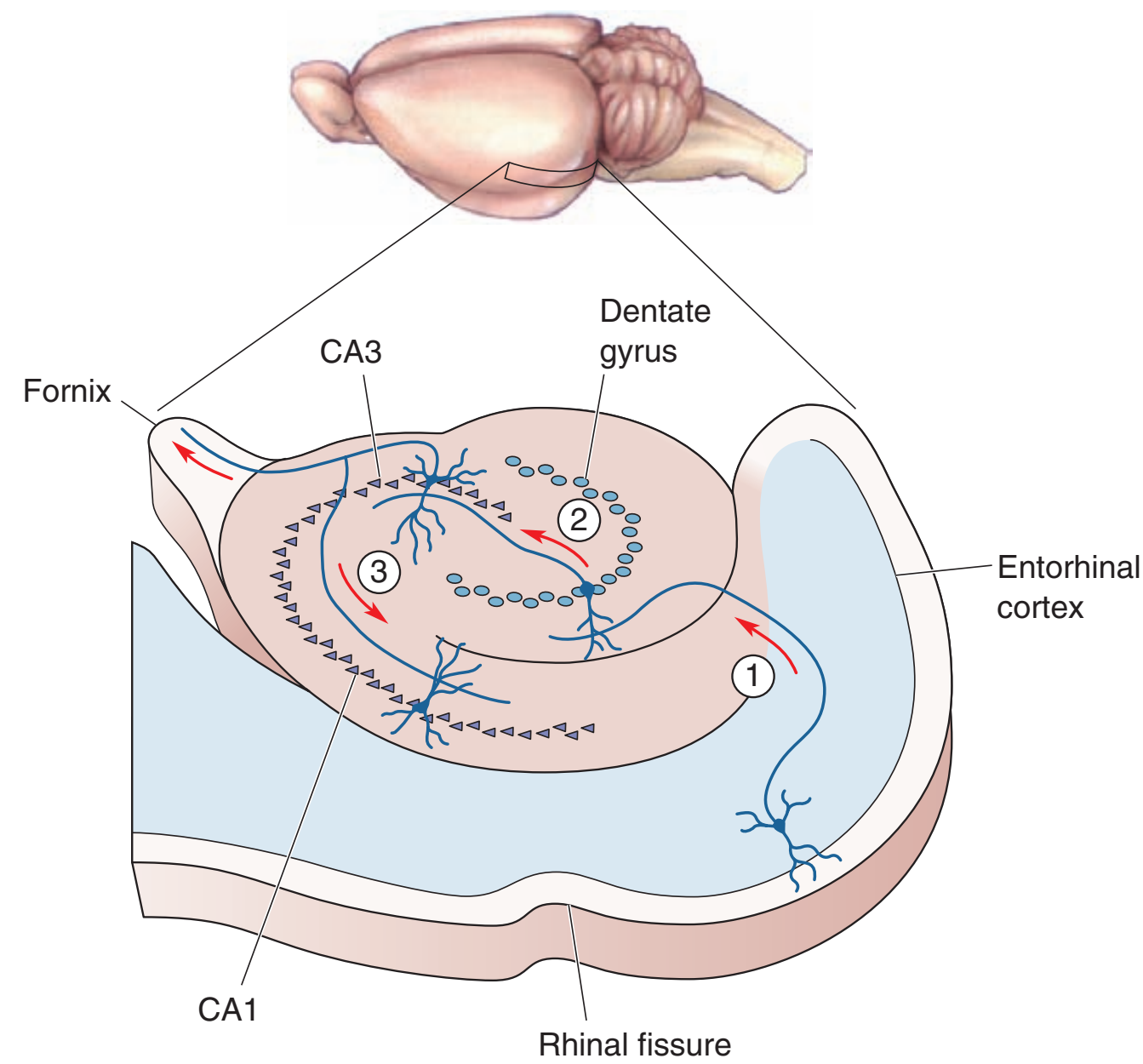
EPSP magnitude in response to test stimulation of input 1



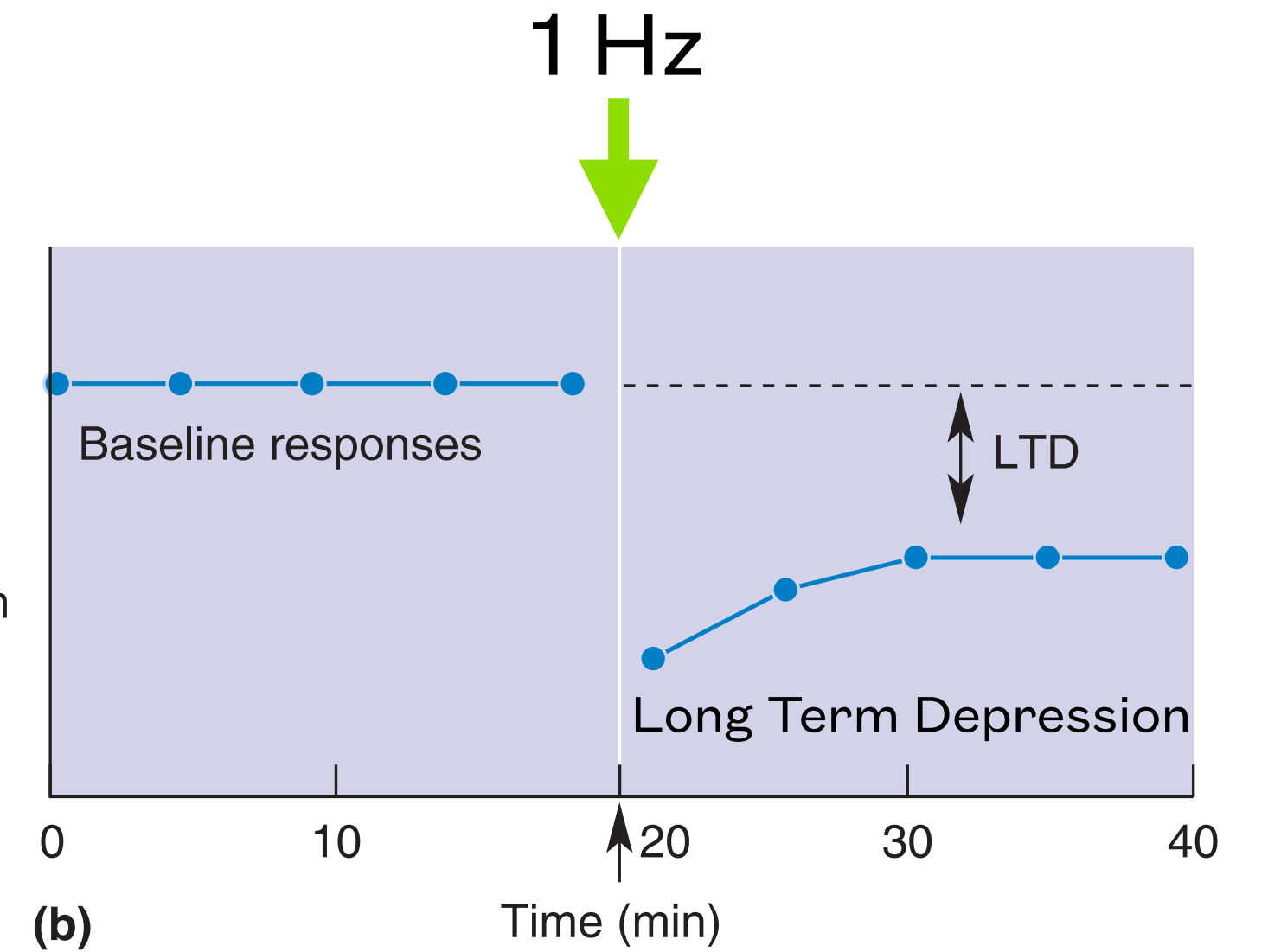
EPSP magnitude in response to test stimulation of input 2



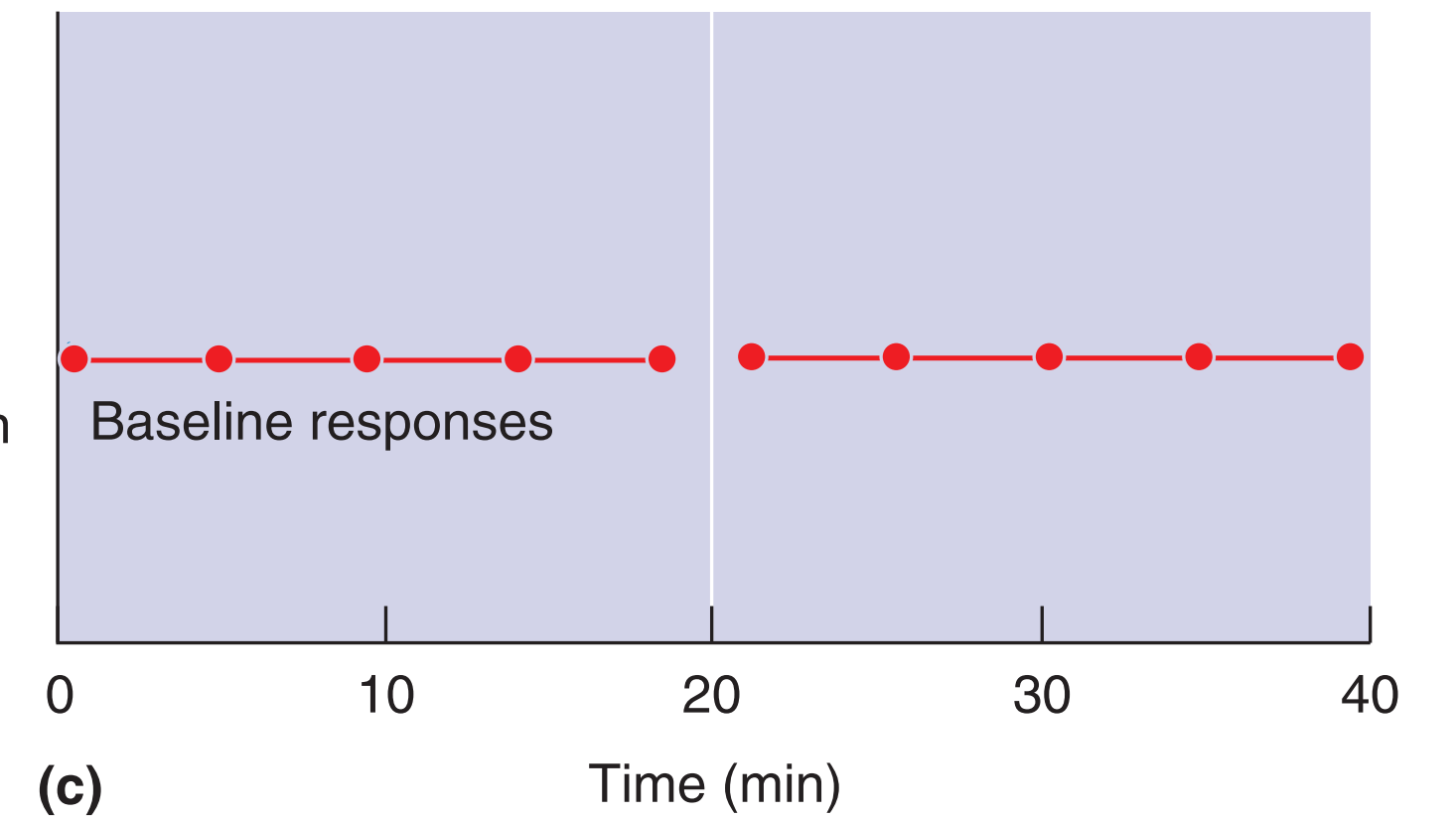
LTD in Hippocampus



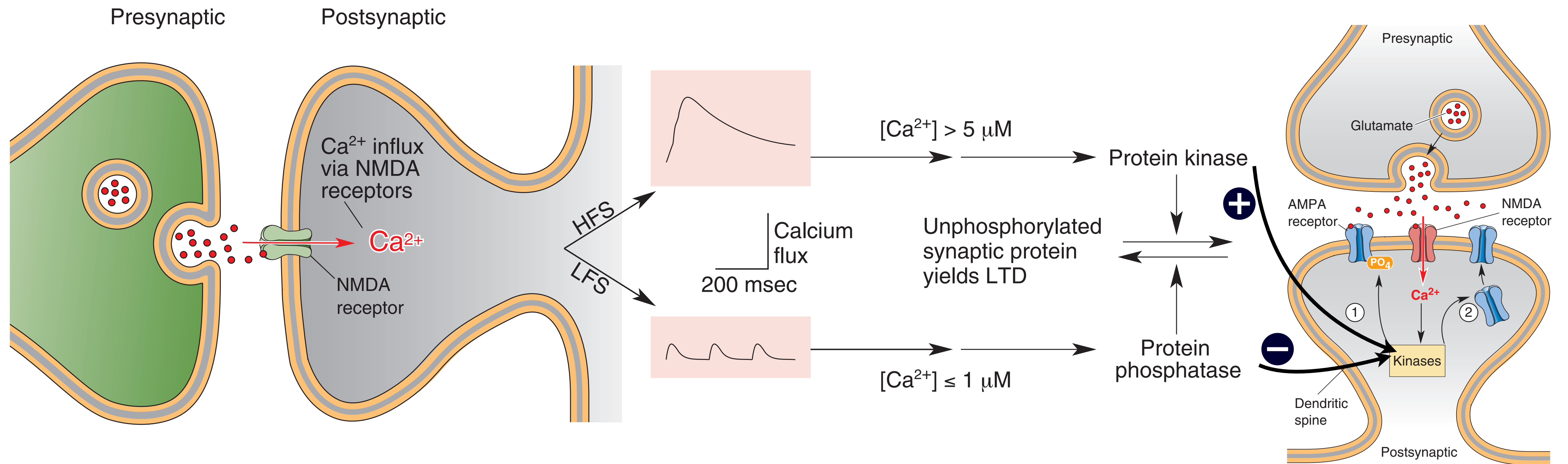
EPSP magnitude in response to test stimulation of input 1



EPSP magnitude in response to test stimulation of input 2

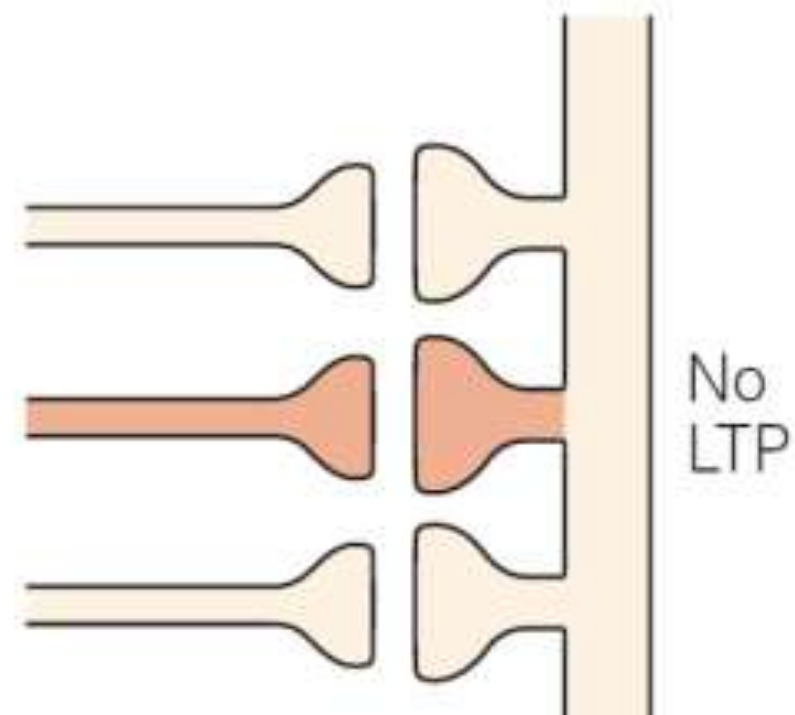


LTP vs. LTD

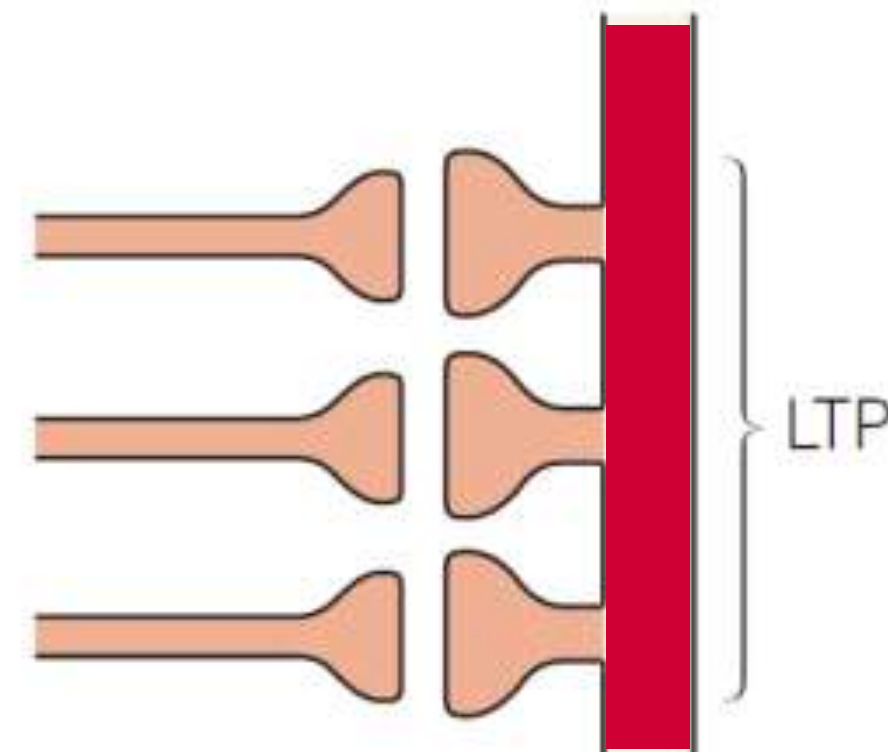


LTP does many things

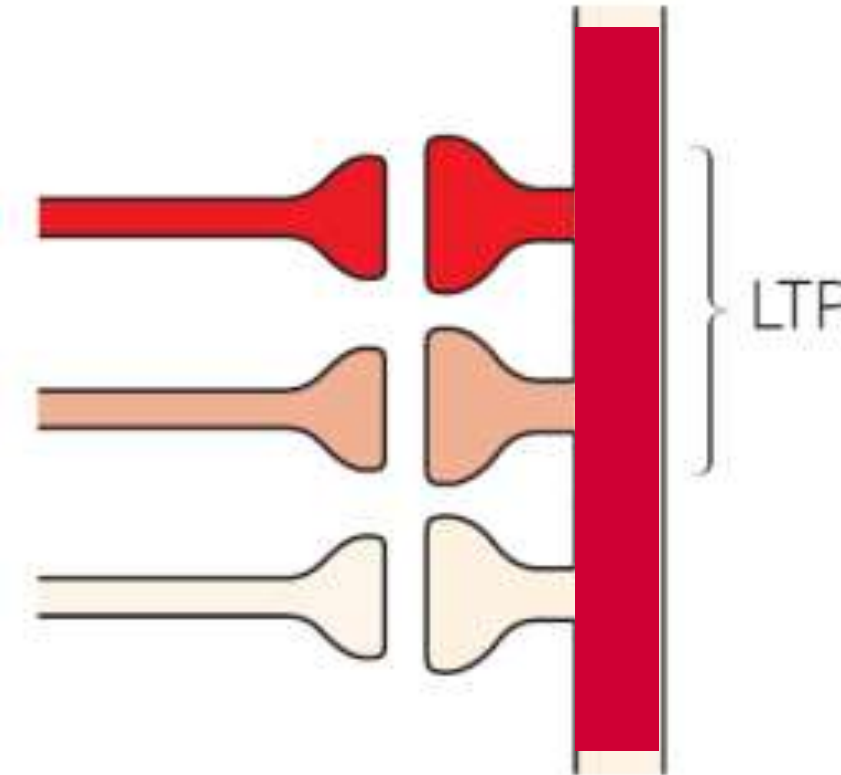
Normal synaptic transmission



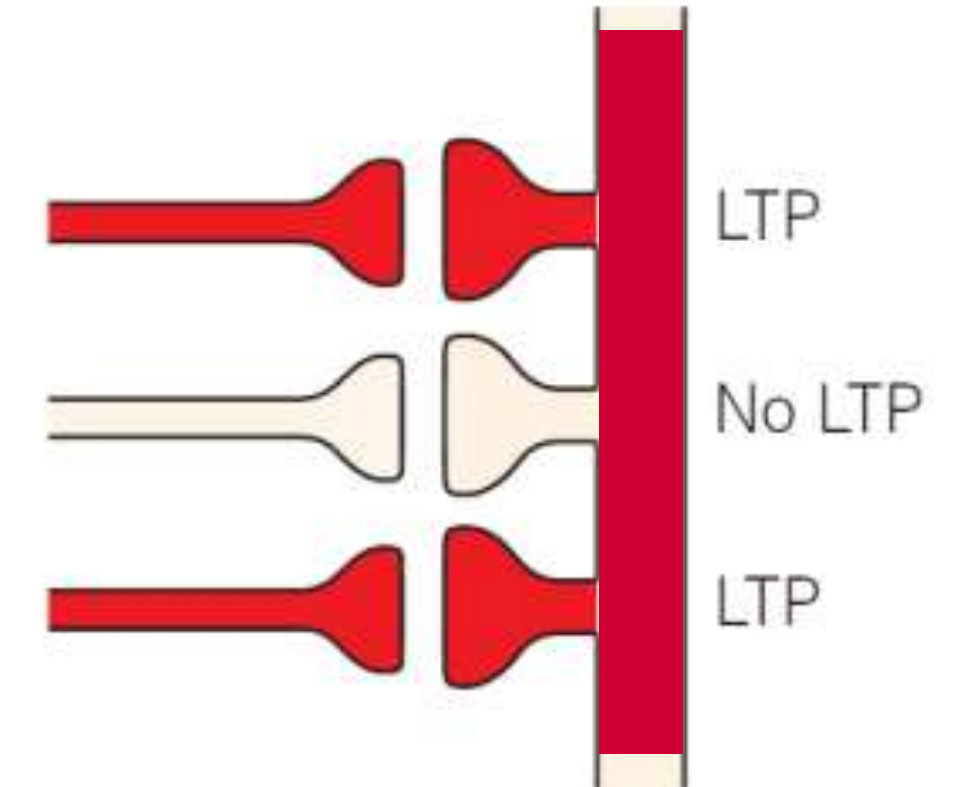
Cooperativity



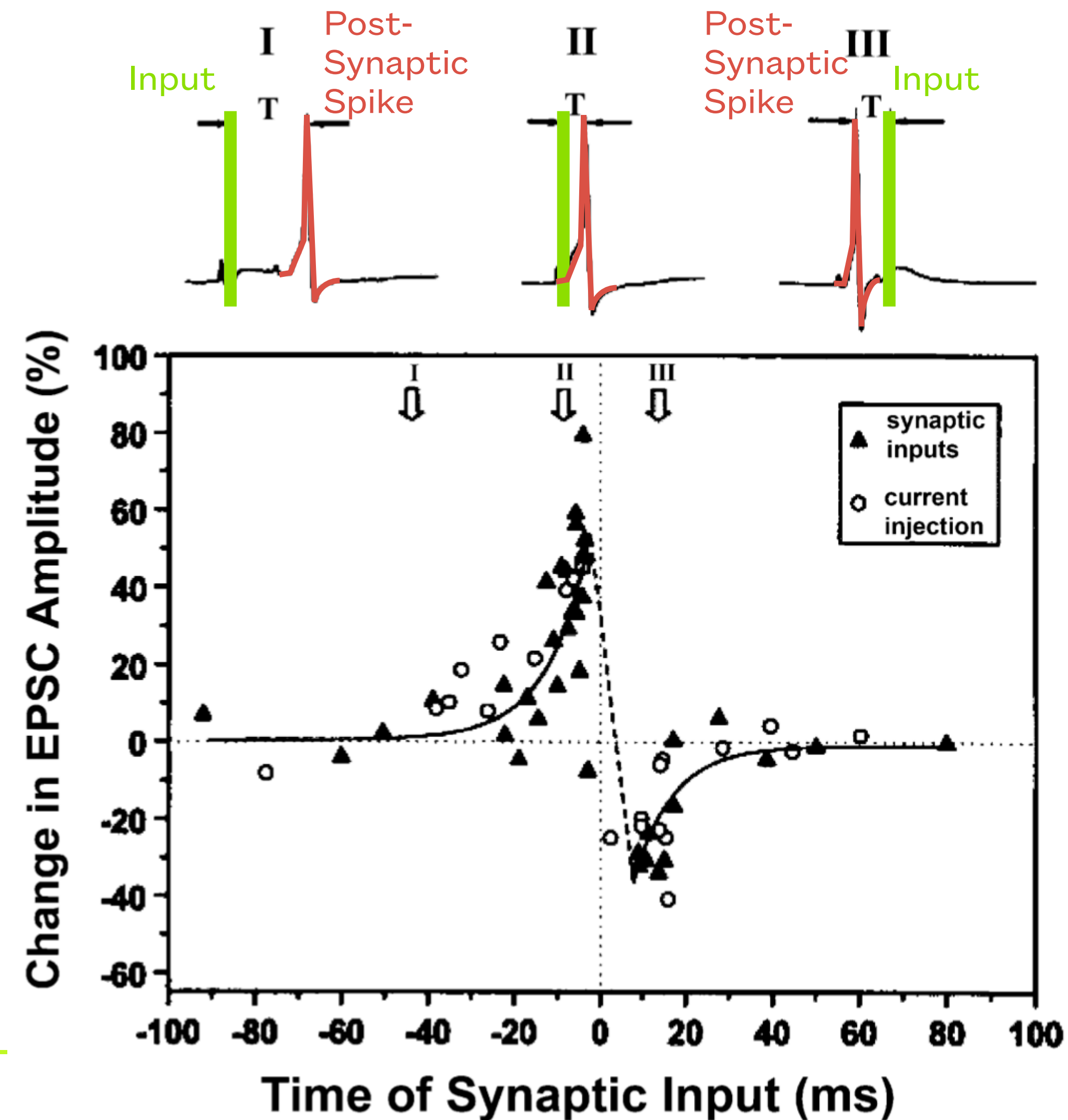
Associativity



Synapse specificity

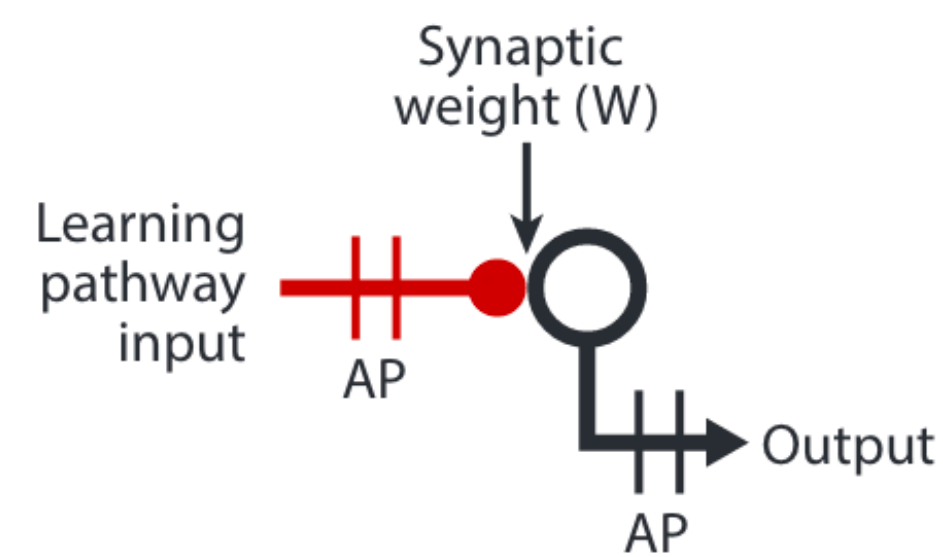


Spike Time Dependent Plasticity



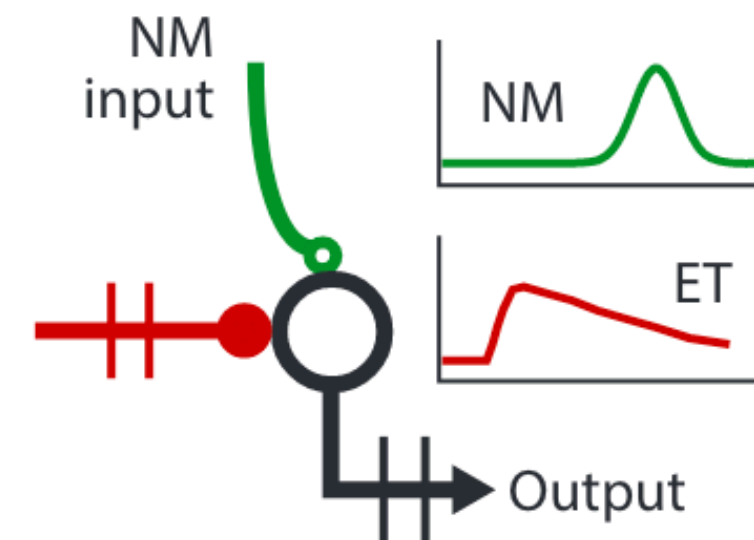
Extensions of STDP

a Hebbian STDP



$$\Delta W = \text{input} \times \text{output}$$

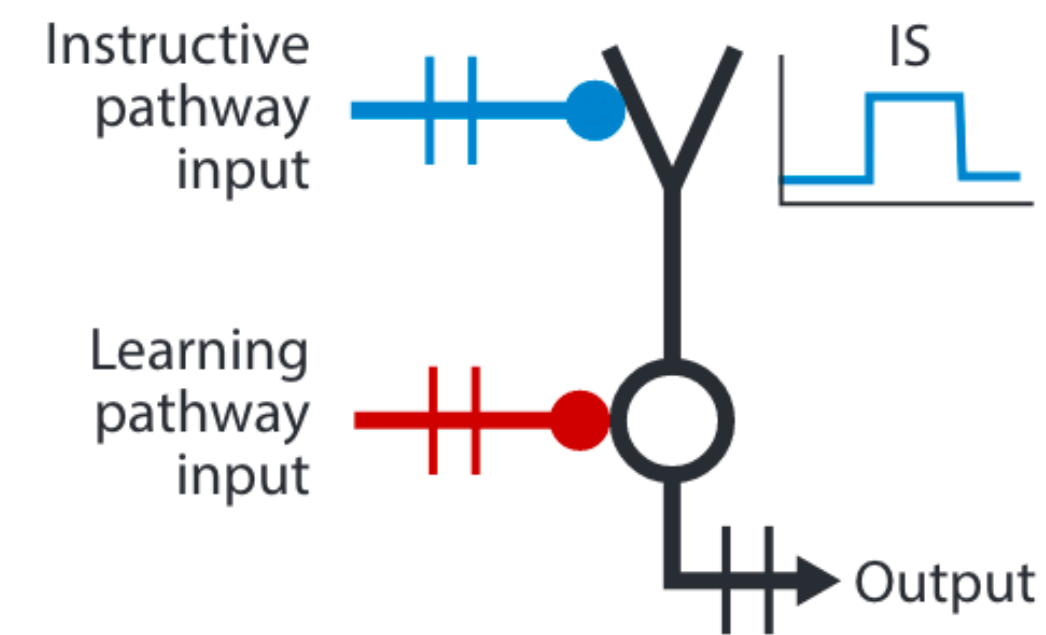
b Neuromodulated STDP



$$\begin{aligned} \text{NM} &= \text{novelty, reward, etc.} \\ \text{ET}_{(t+1)} &= (\text{input} \times \text{output}) - \text{ET}_{(t)}/\tau_e \\ \Delta W &= \text{ET} \times \text{NM} \end{aligned}$$

ET: Eligibility Trace

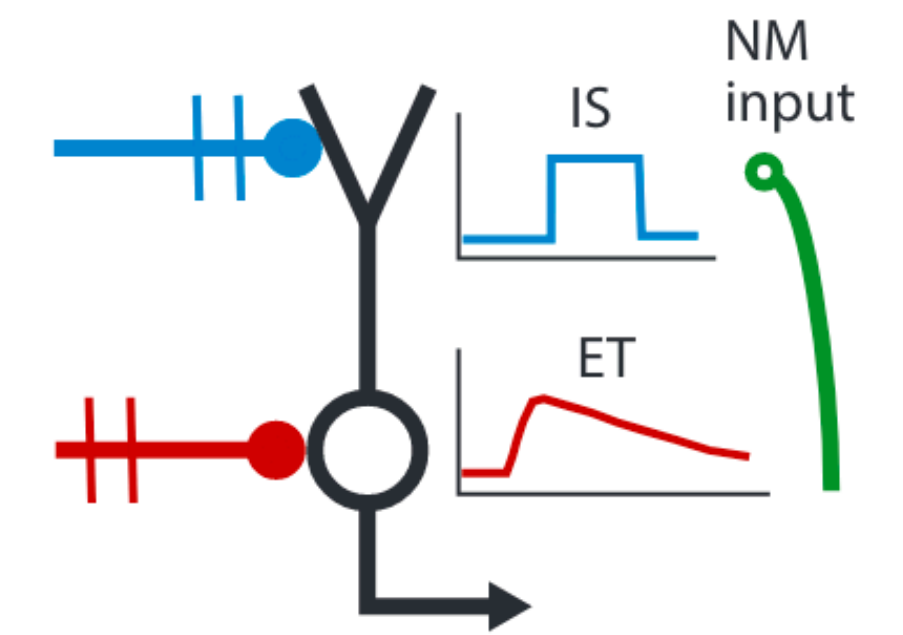
c Supervised



$$\begin{aligned} \text{IS} &= \text{desired} - \text{actual} \\ \Delta W &= \text{input} \times \text{IS} \end{aligned}$$

IS: Instruction Signal

d BTSP



$$\begin{aligned} \text{IS} &= \text{local error} \cdot \text{NM} \\ \text{ET}_{(t+1)} &= \text{input} - \text{ET}_{(t)}/\tau_e \\ \Delta W &= \text{ET} \times \text{IS} \end{aligned}$$

Contributions from Chinese



冯德培

Feng TP. (1941) The changes in the endplate potential during and after prolonged stimulation. Chin. J. Physiol. 13:79-107



张香桐

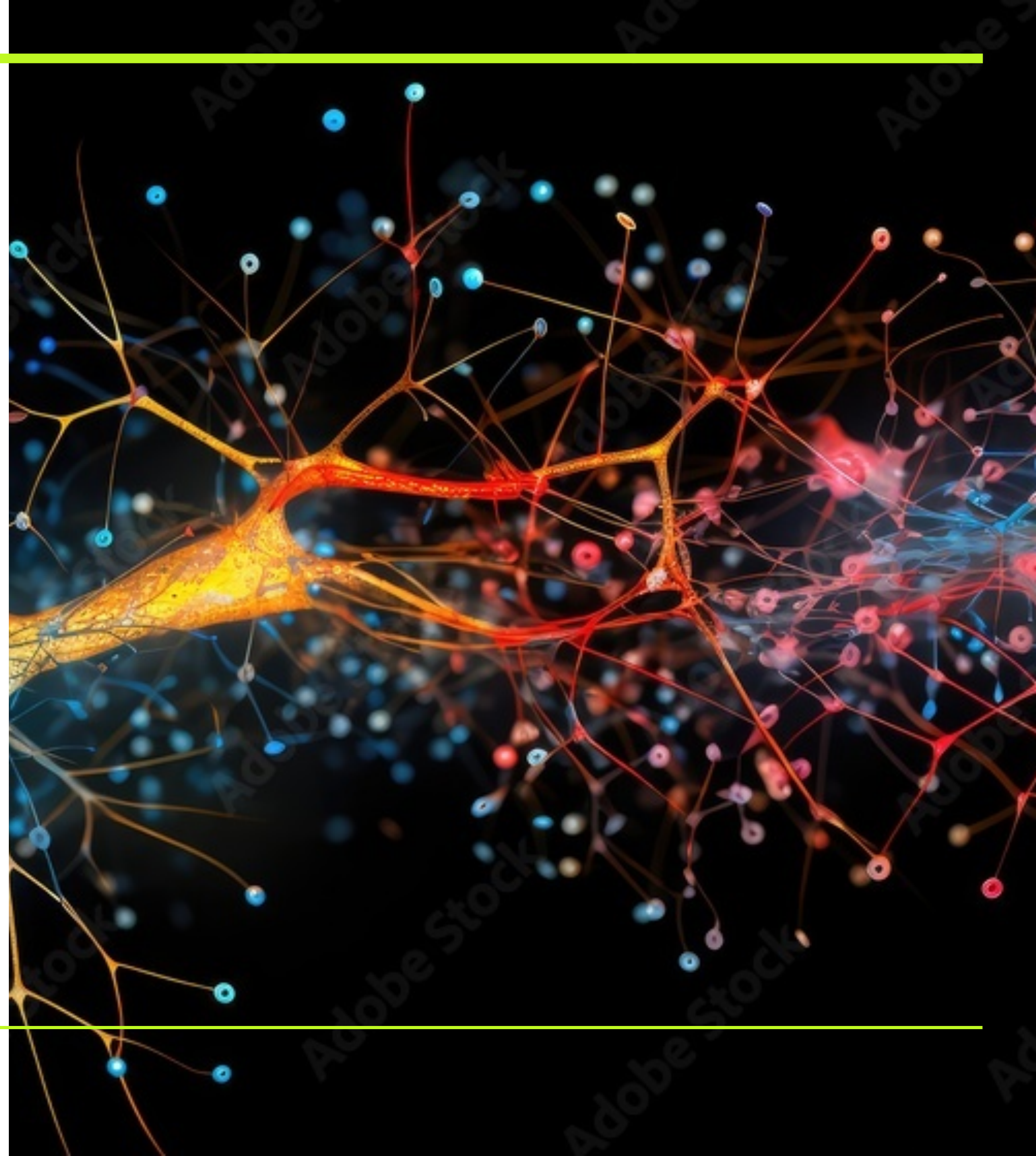
Cheng, HT. (1951) Dendritic potential of cortical neurons produced by direct electrical stimulation of the cerebral cortex. J Neurophysiol. 14(1):1-21.



蒲慕明

Bi and Poo. (1998) Synaptic modifications in cultured hippocampal neurons: dependence on spike timing, synaptic strength, and postsynaptic cell type. J Neurosci. 18(24):10464-72

-
- ❖ Overview
 - ❖ Synaptic Mechanism
 - ❖ **Hebbian Learning**
 - ❖ Hopfield Network
-



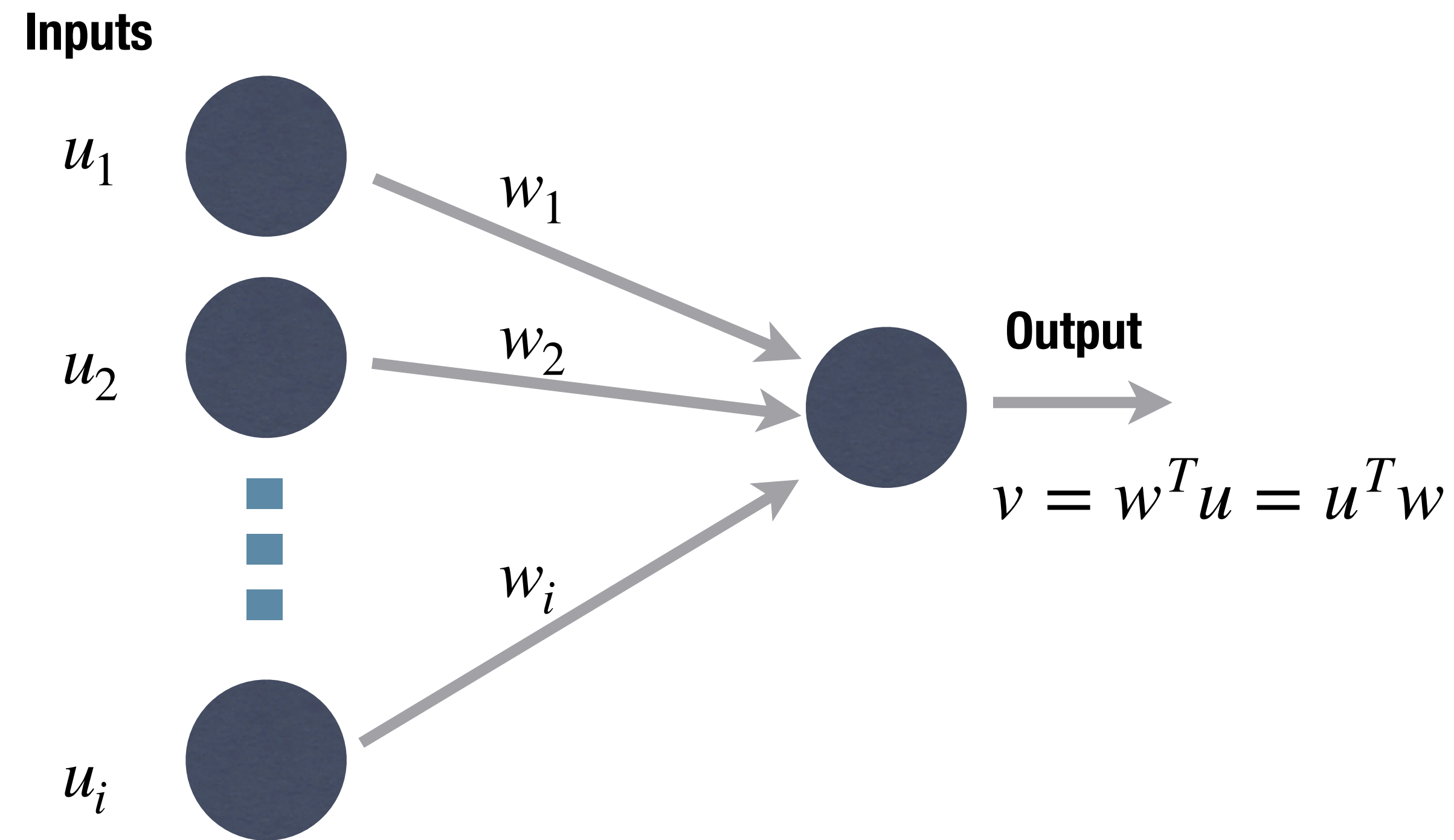
Hebb rule

**NEURONS THAT
FIRE TOGETHER
WIRE TOGETHER**



Donald Hebb
1904-1985

Hebbian learning



$$\tau_w \frac{dw}{dt} = uv$$

$$\tau_w \frac{dw}{dt} = \langle uv \rangle_u \text{ Average over all inputs}$$

$$\tau_w \frac{dw}{dt} = \langle uv \rangle = \langle uu^T w \rangle = \langle uu^T \rangle w$$

Input Correlation

Hebbian learning

Correlation Rule

$$\tau_w \frac{dw}{dt} = uv$$

$$\tau_w \frac{dw}{dt} = Q \bullet w$$

$$Q = \langle uu^T \rangle$$

Covariance Rule

$$\tau_w \frac{dw}{dt} = u(v - \theta_v) \quad \tau_w \frac{dw}{dt} = (u - \theta_u)v$$

$$\theta_u = \langle u \rangle$$

$$\tau_w \frac{dw}{dt} = C \bullet w$$

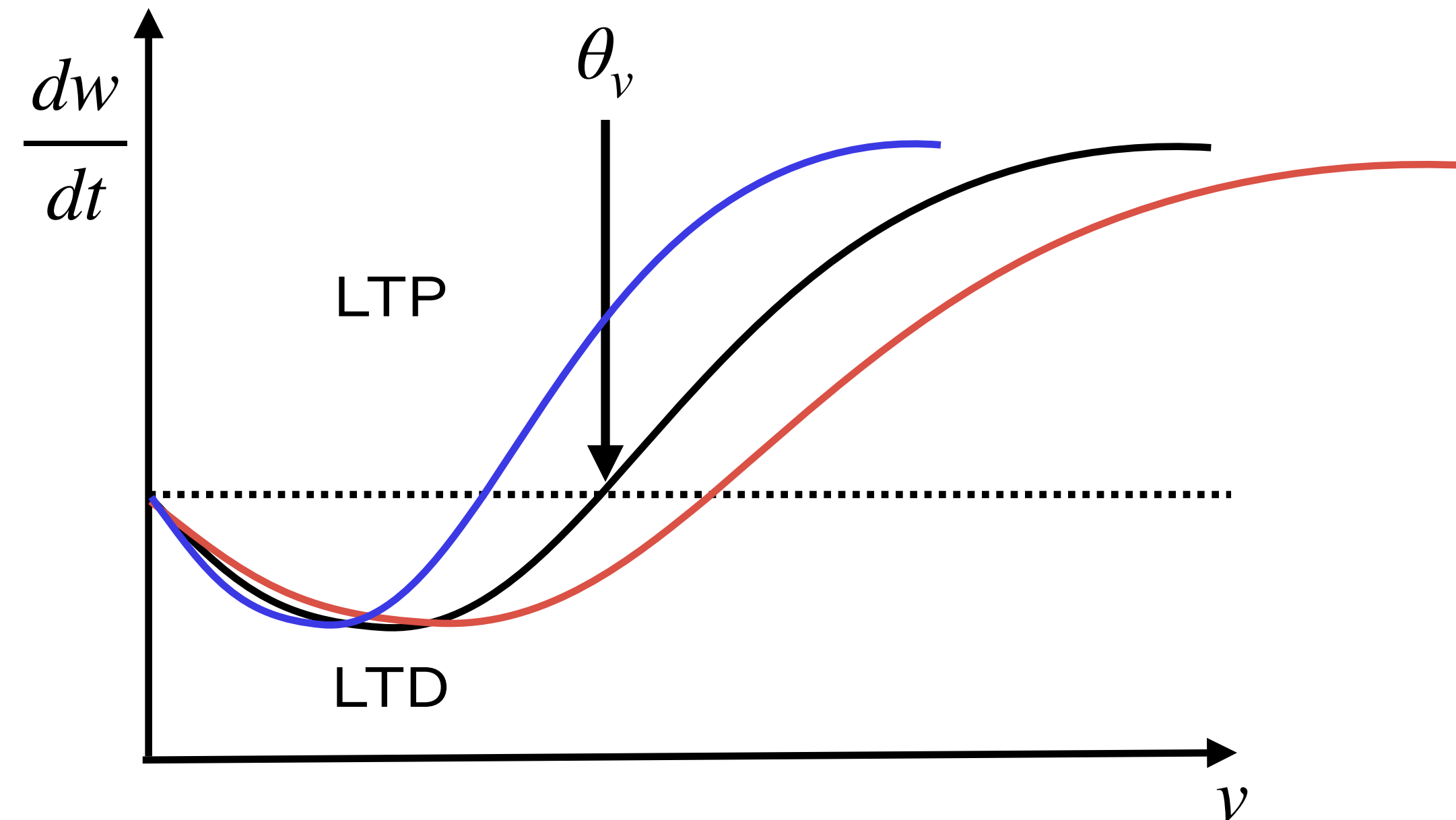
$$C = (\langle u - \langle u \rangle \rangle)(\langle u - \langle u \rangle \rangle)^T$$

BCM Rule

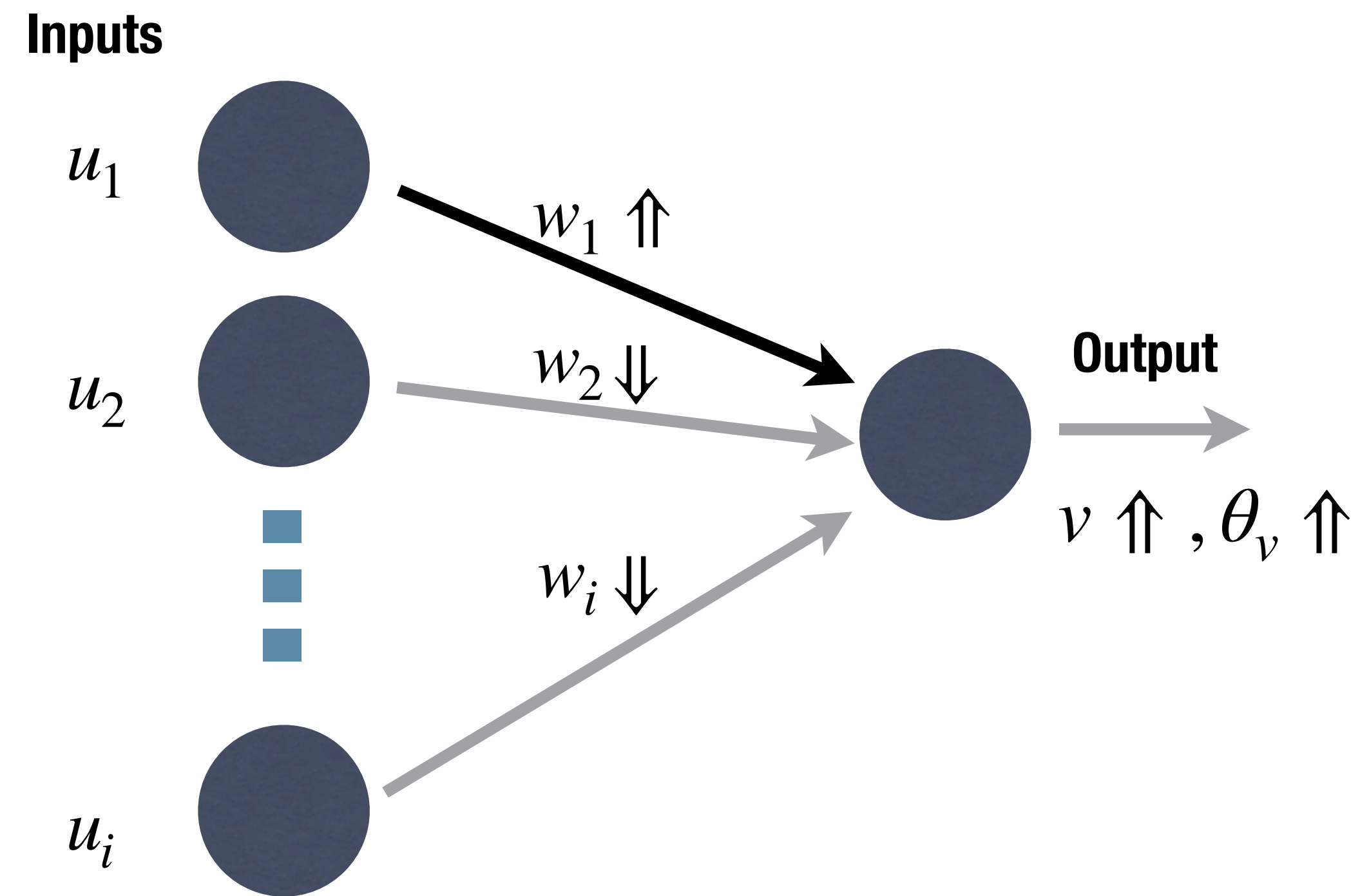
BCM Rule

$$\tau_w \frac{dw}{dt} = u(v - \theta_v)v$$
$$\tau_\theta \frac{d\theta_v}{dt} = v^2 - \theta_v$$

$$\tau_\theta < \tau_w$$



BCM Implements Synaptic Competition

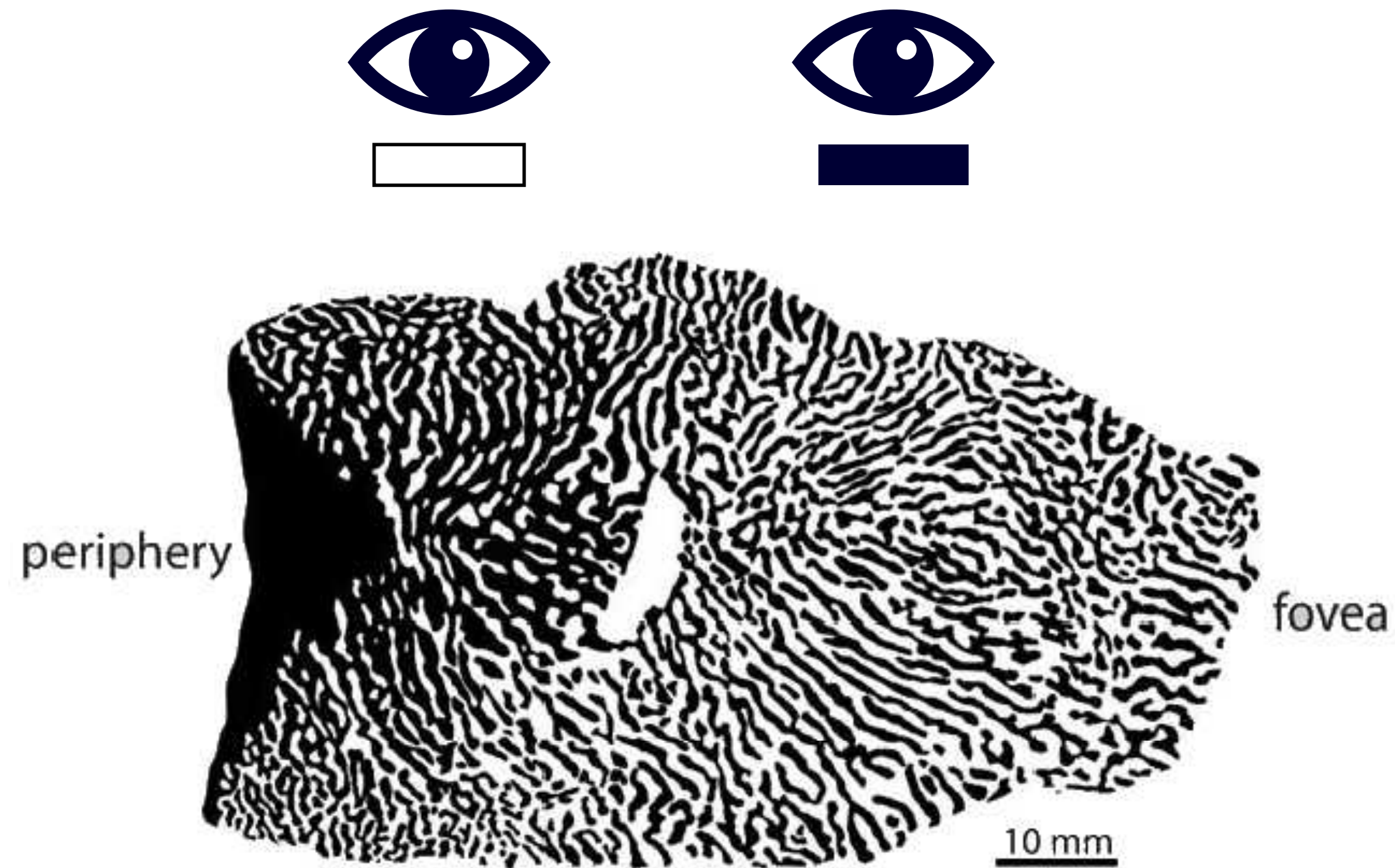


BCM Rule

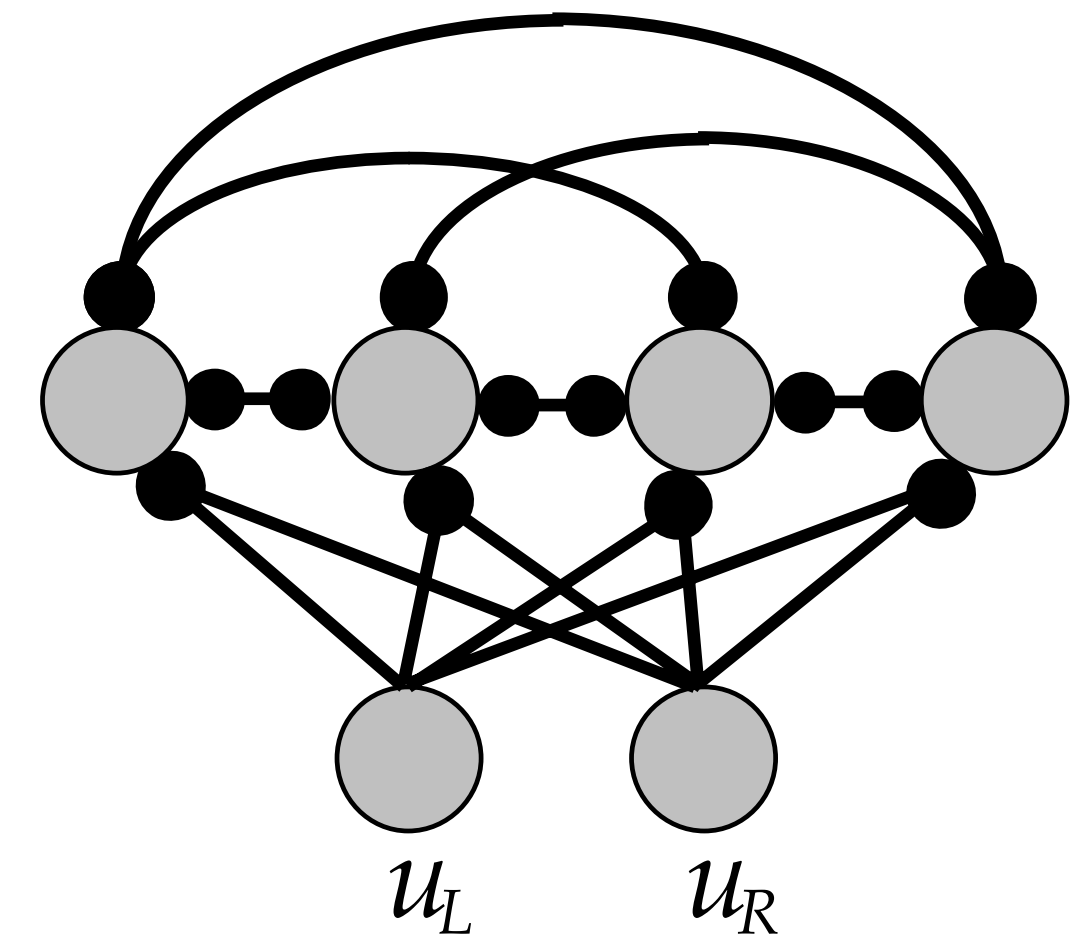
$$\tau_w \frac{dw}{dt} = u(v - \theta_v)v$$

$$\tau_\theta \frac{d\theta_v}{dt} = v^2 - \theta_v$$

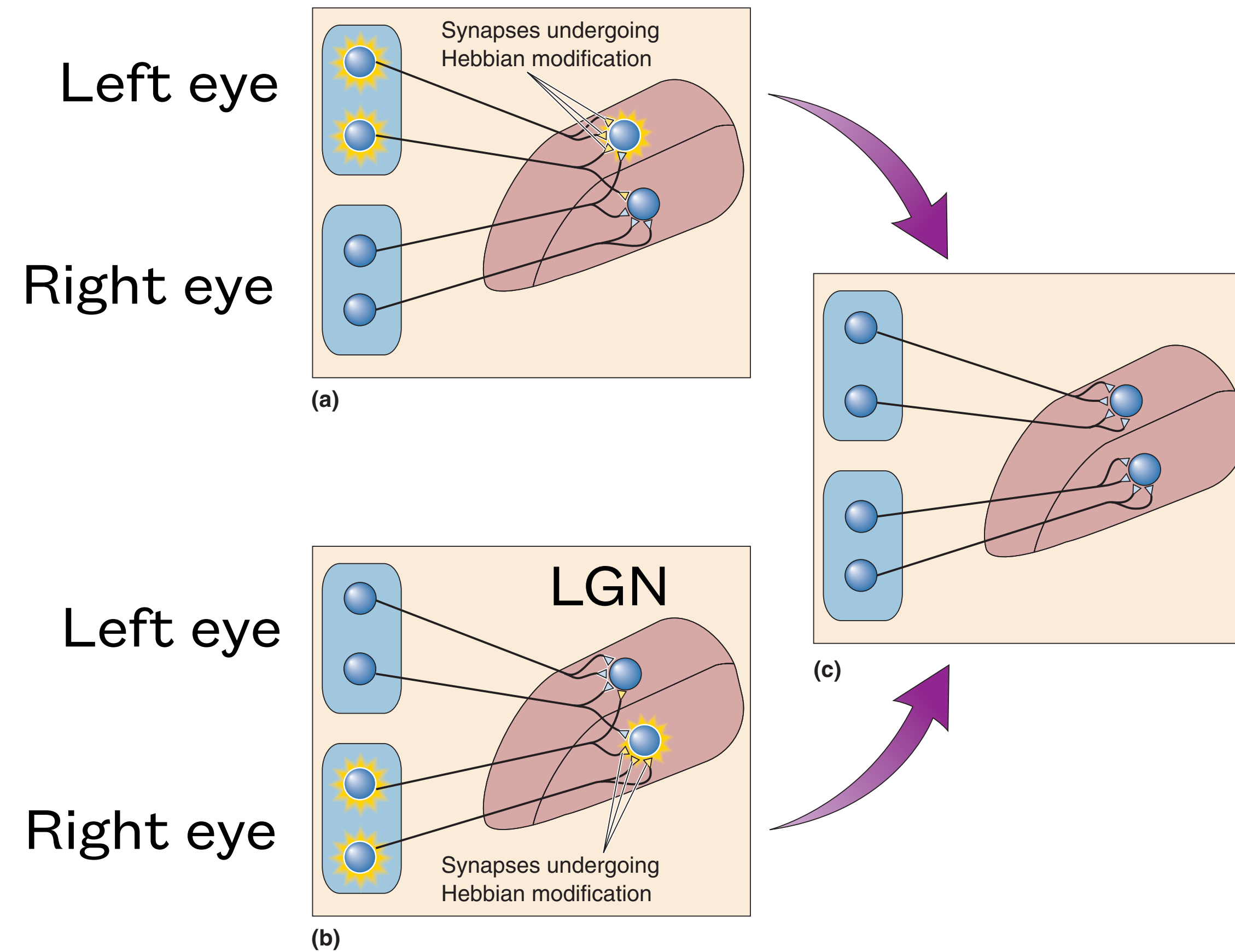
Hebbian learning predicts ocular dominance columns



Adams et al., 2007

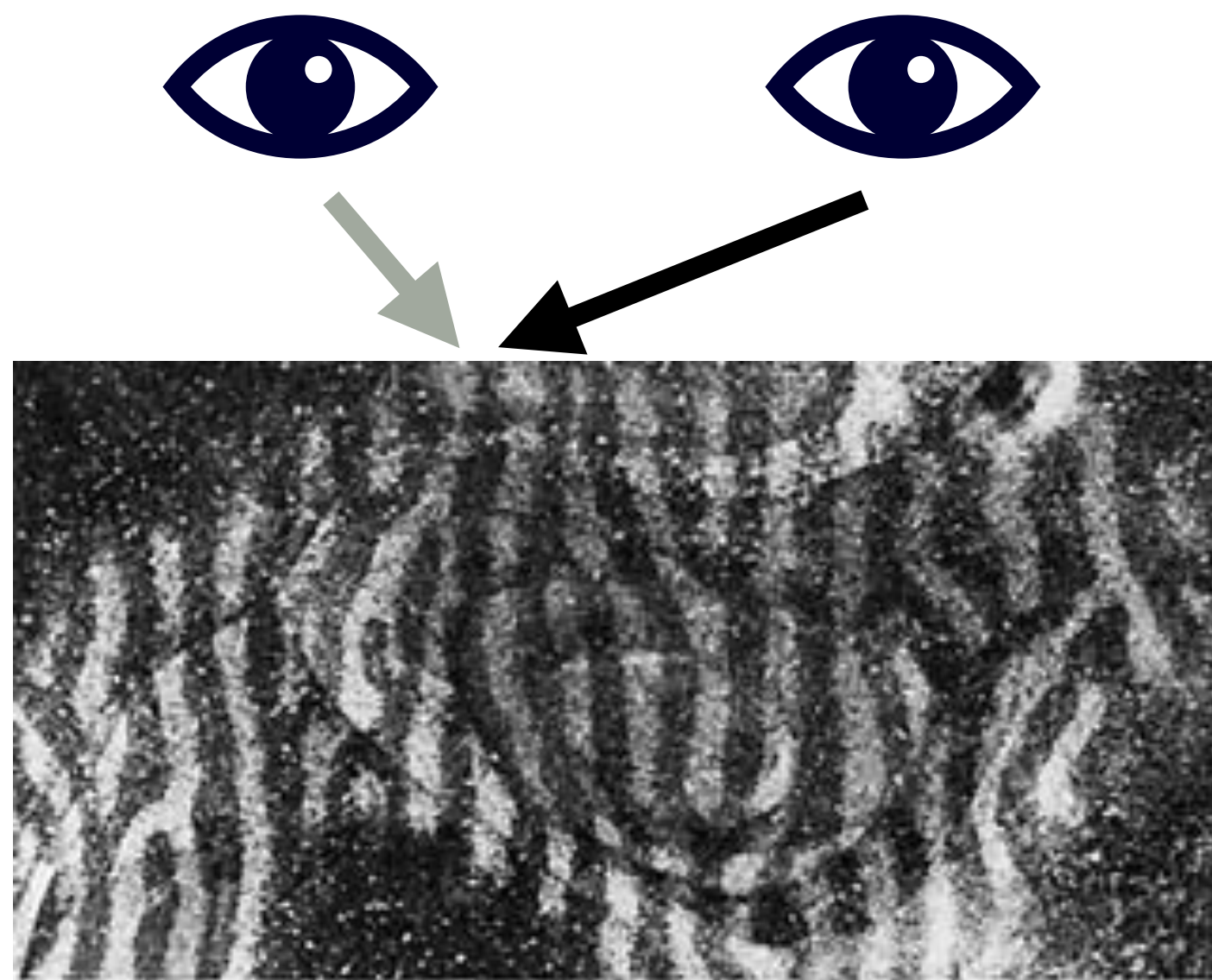


Synaptic Segregation

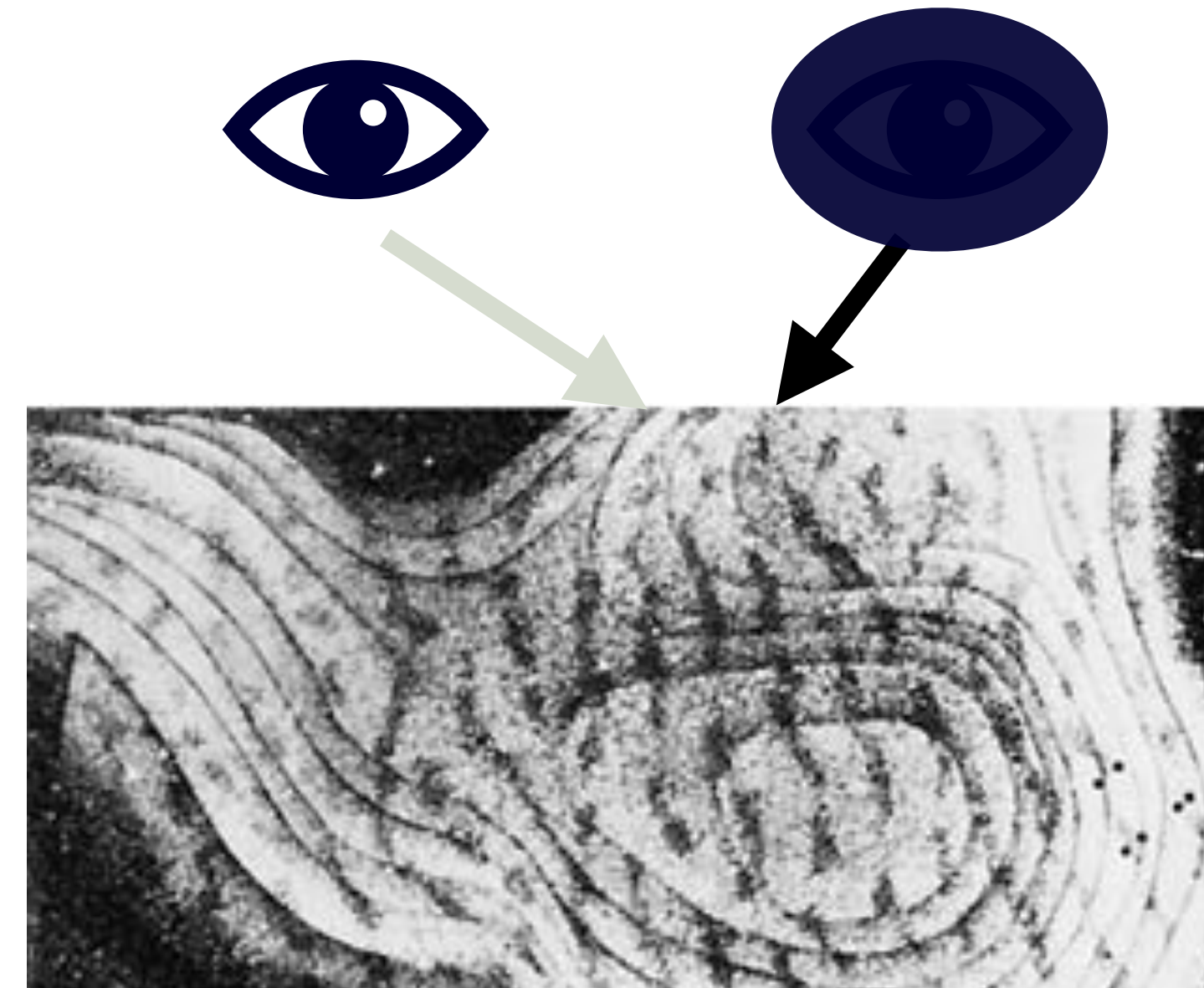


Synaptic Segregation

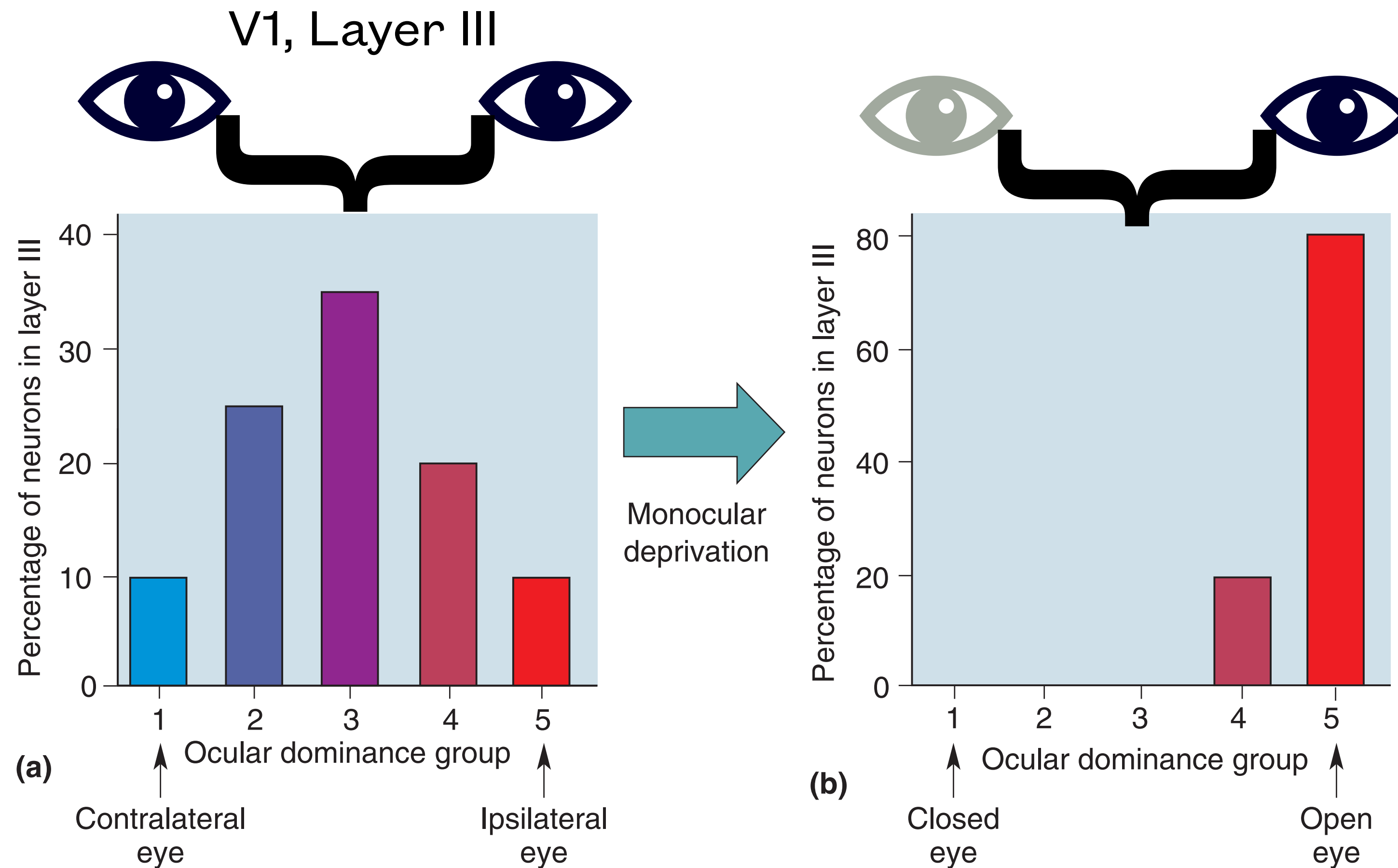
22 month, starting from week 2



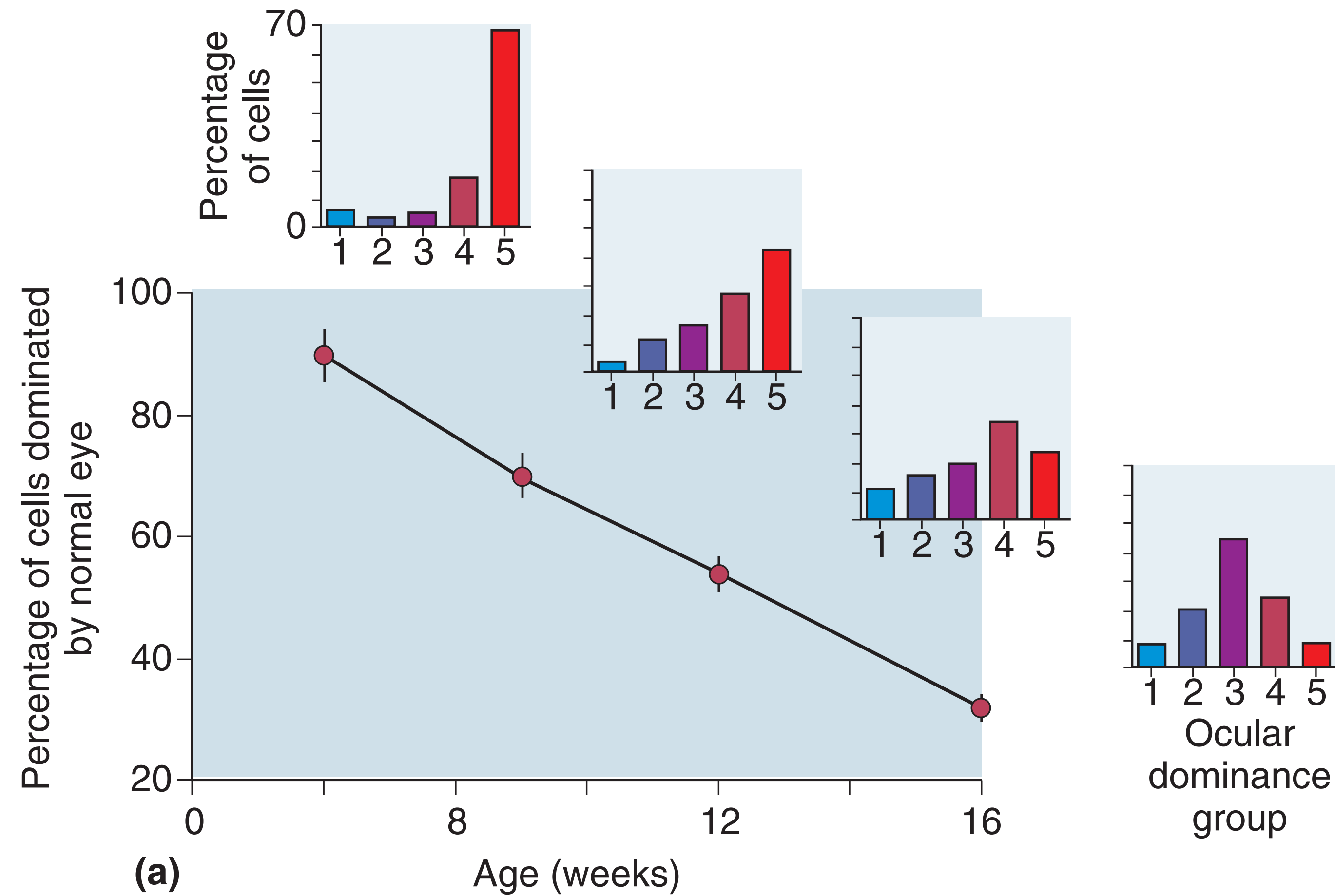
Layer IV, V1



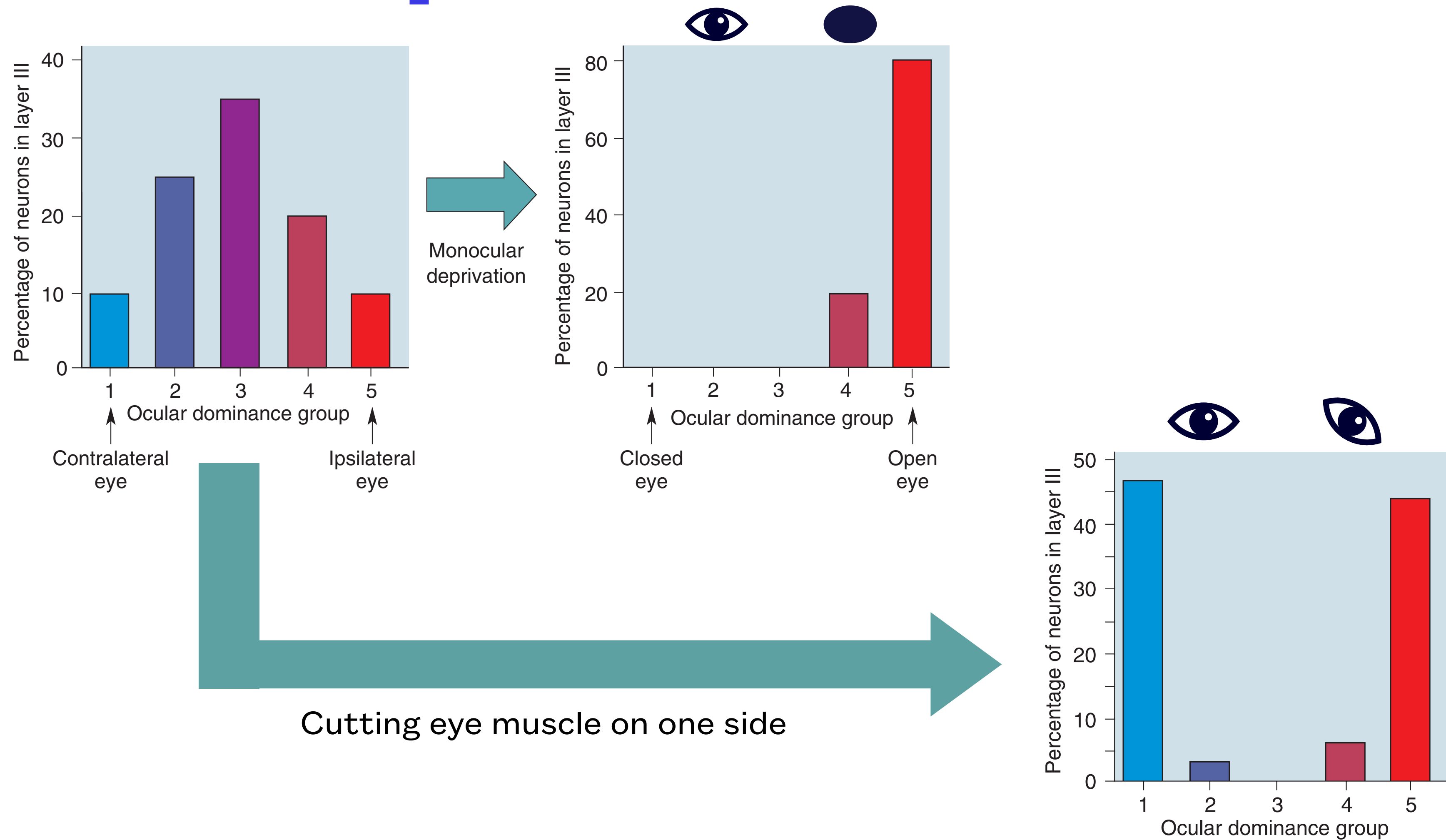
Synaptic Convergence



Critical Period



Synaptic Competition



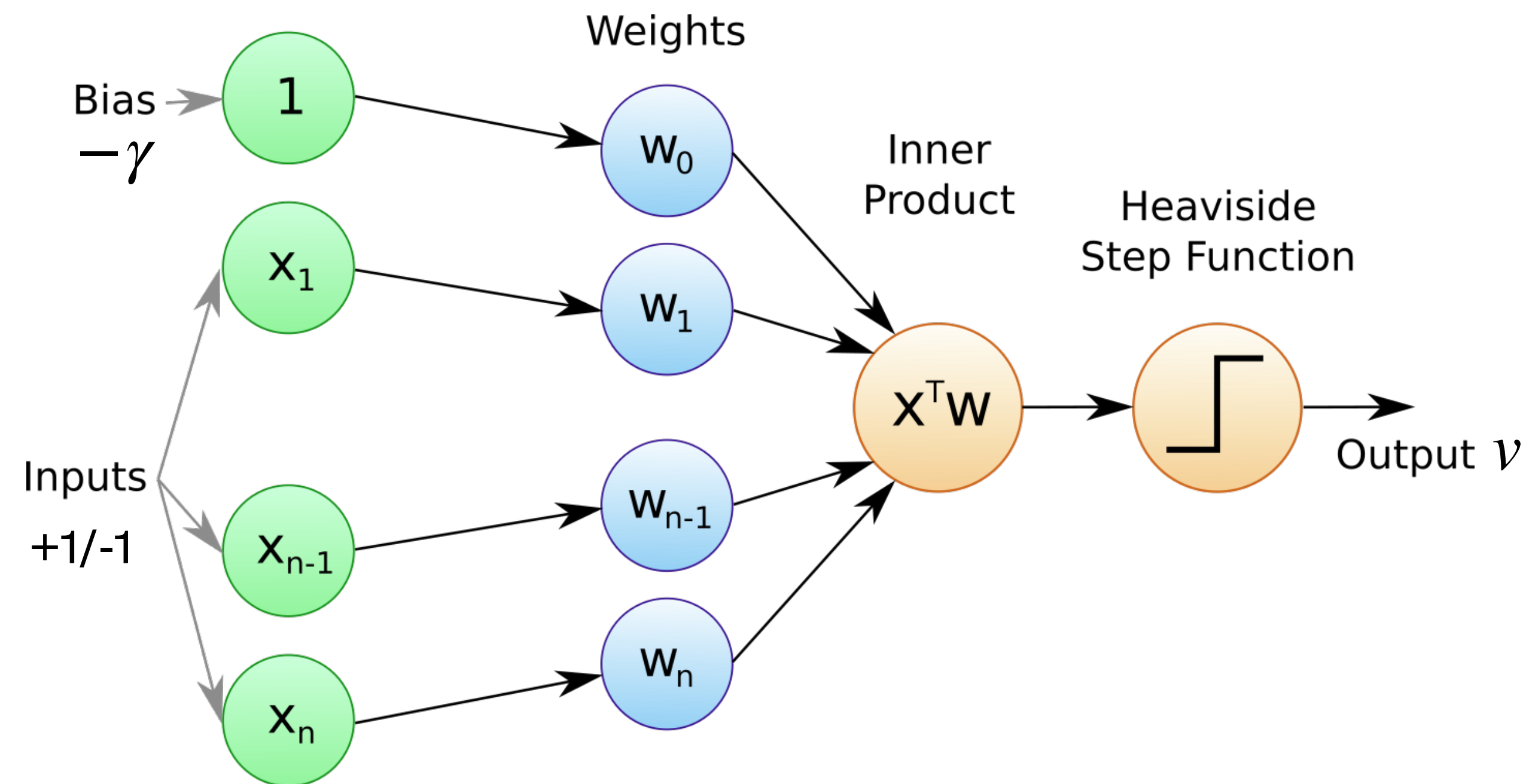
Supervised Hebbian Learning

Input-Output pairs: $\begin{cases} u^m \\ v^m \end{cases} \quad m = 1..N_s$

$$\tau_w \frac{dw}{dt} = \langle uv \rangle = \frac{1}{N_s} \sum_{m=1}^{N_s} v^m u^m \quad \text{Never stops!}$$

$$\tau_w \frac{dw}{dt} = \langle uv \rangle - \alpha w \quad w = \langle uv \rangle / \alpha$$

Perceptron



$$v = \begin{cases} +1 & \text{if } wx - \gamma \geq 0 \\ -1 & \text{if } wx - \gamma < 0 \end{cases}$$

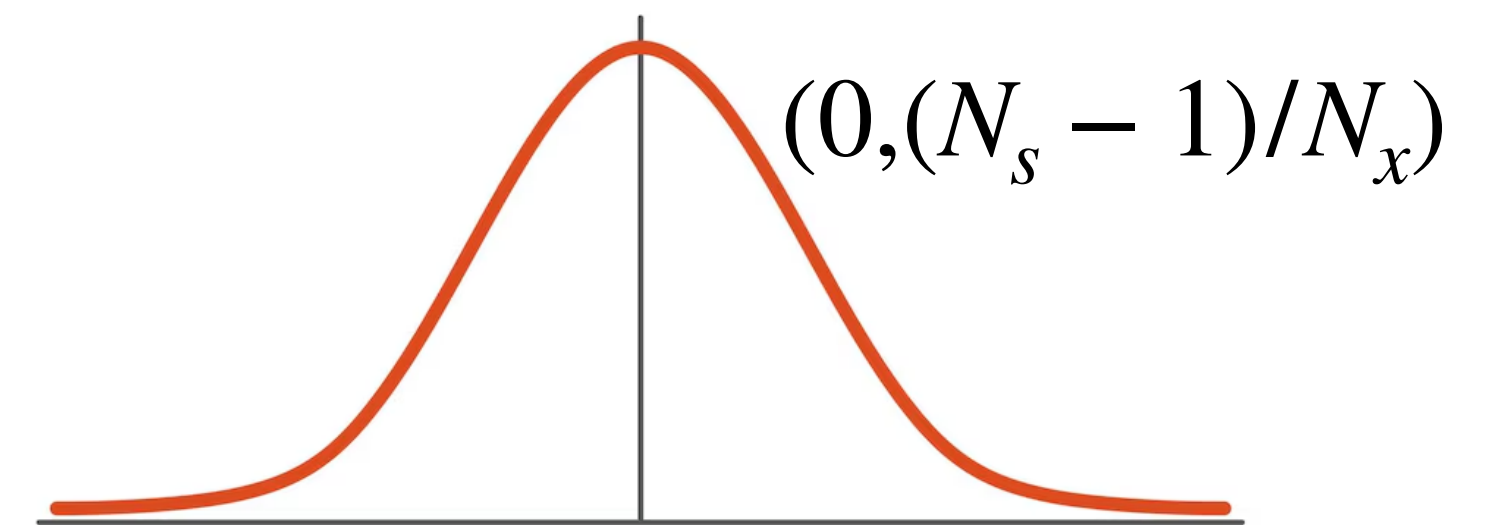
$$w = \langle xv \rangle / \alpha = \frac{1}{N_x} \sum_{m=1}^{N_s} v^m x^m$$

$$\alpha = N_x / N_s$$

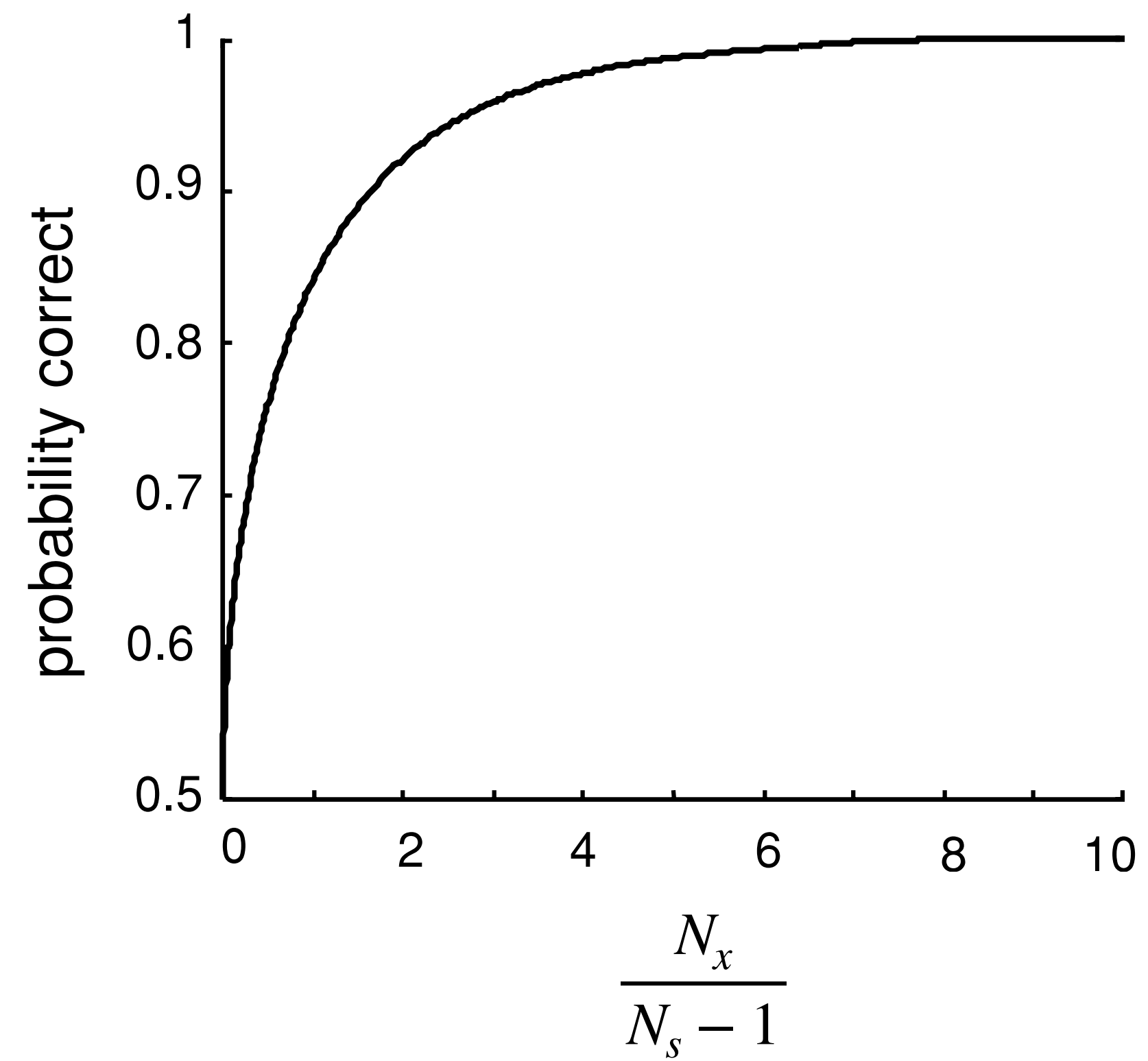
$$\gamma = 0$$

$$wx^i = \underbrace{\frac{1}{N_x} v_i x_i \cdot x^i}_{v^i} + \underbrace{\frac{1}{N_x} \sum_{m \neq i} v_m x_m \cdot x^i}_{\text{Interference } N_s - 1}$$

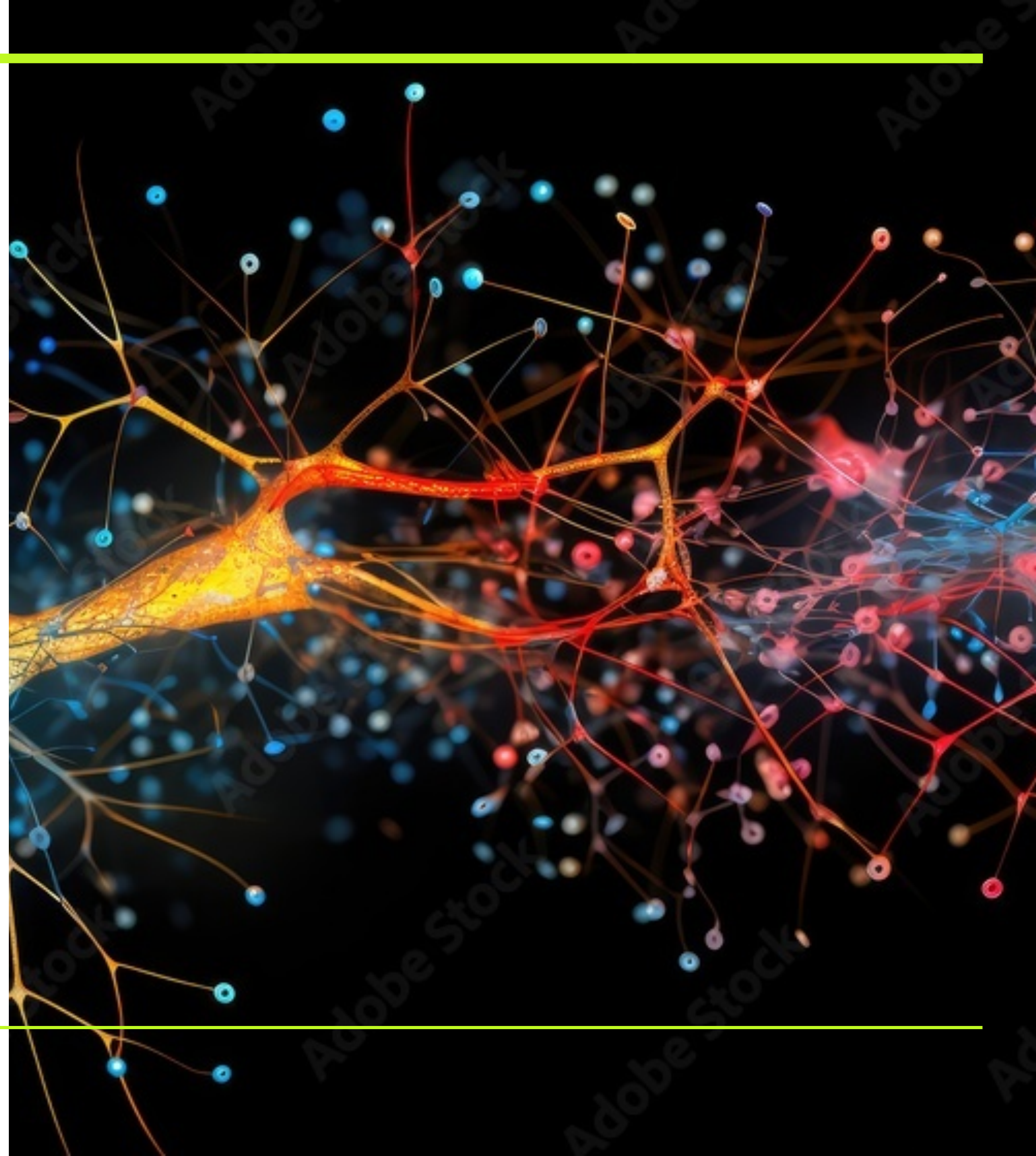
$$wx^i = v^i + \eta^i$$



Perceptron



-
- ❖ Overview
 - ❖ Synaptic Mechanism
 - ❖ Hebbian Learning
 - ❖ **Hopfield Network**



Hopfield network

Associative memory:

Recall whole pattern associated with partial information

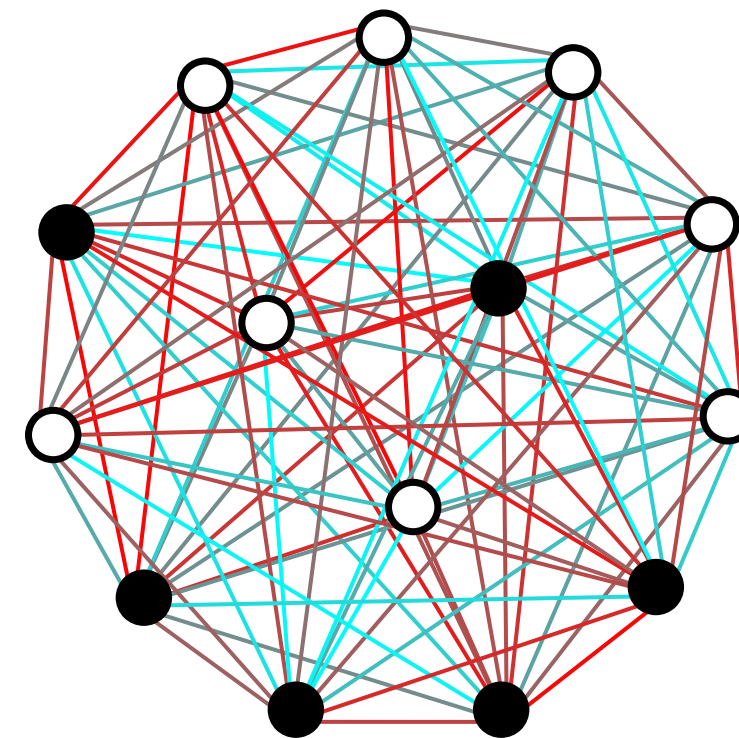
Recurrent network with no explicit readout.

Internal neural activity is the target.

N binary 'neurons': $s_i = \pm 1$

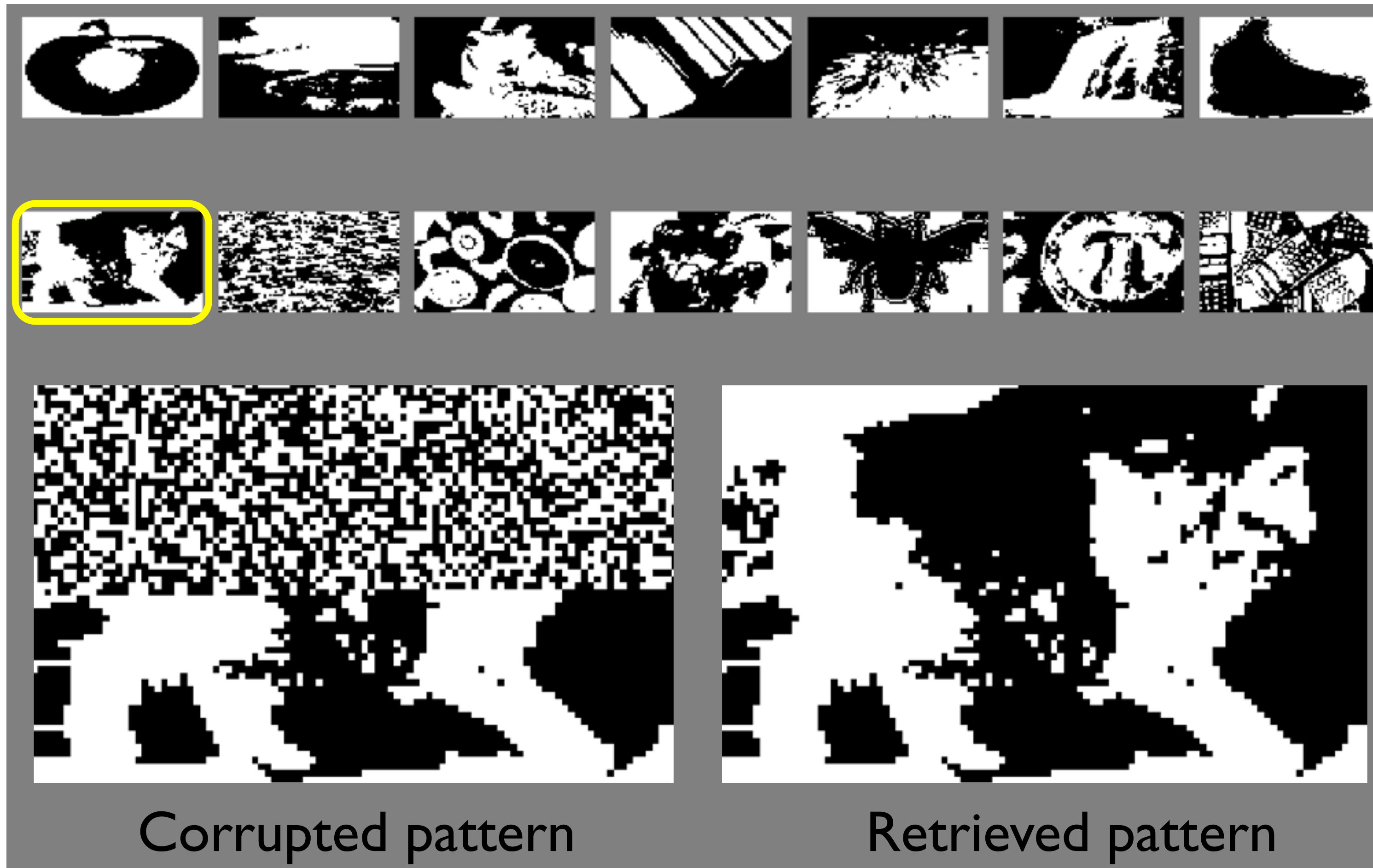
p input patterns: $\xi_i^\mu = \pm 1$

N^2 synapses: $W_{ij} = W_{ji}, W_{ii} = 0$



John Hopfield
2024 Nobel Physics

Hopfield network

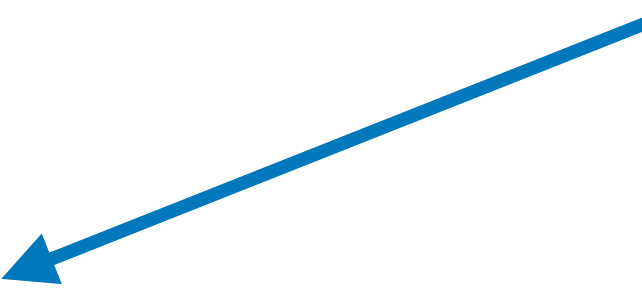


Hopfield network

Dynamics: $s_i \leftarrow \text{sgn}(\sum_j W_{ij}s_j - \theta_i)$

Learning rule: $W_{ij} \leftarrow W_{ij} + \xi_i^\mu \xi_j^\mu$

Hebb rule!



$$W_{ij} = \sum_{\mu} \xi_i^\mu \xi_j^\mu$$

i, j neuron index

s neuron state

μ pattern index

ξ pattern

Hopfield network

Dynamics: $s_i \leftarrow \operatorname{sgn}\left(\sum_j W_{ij}s_j - \theta_i\right) = \operatorname{sgn}\left(\sum_{\mu,j} \xi_i^\mu \xi_j^\mu s_j - \theta_i\right) = \operatorname{sgn}\left(\sum_{\mu} \xi_i^\mu \left(\sum_j \xi_j^\mu s_j\right) - \theta_i\right) = \operatorname{sgn}\left(\sum_{\mu} \xi_i^\mu v^\mu - \theta_i\right)$

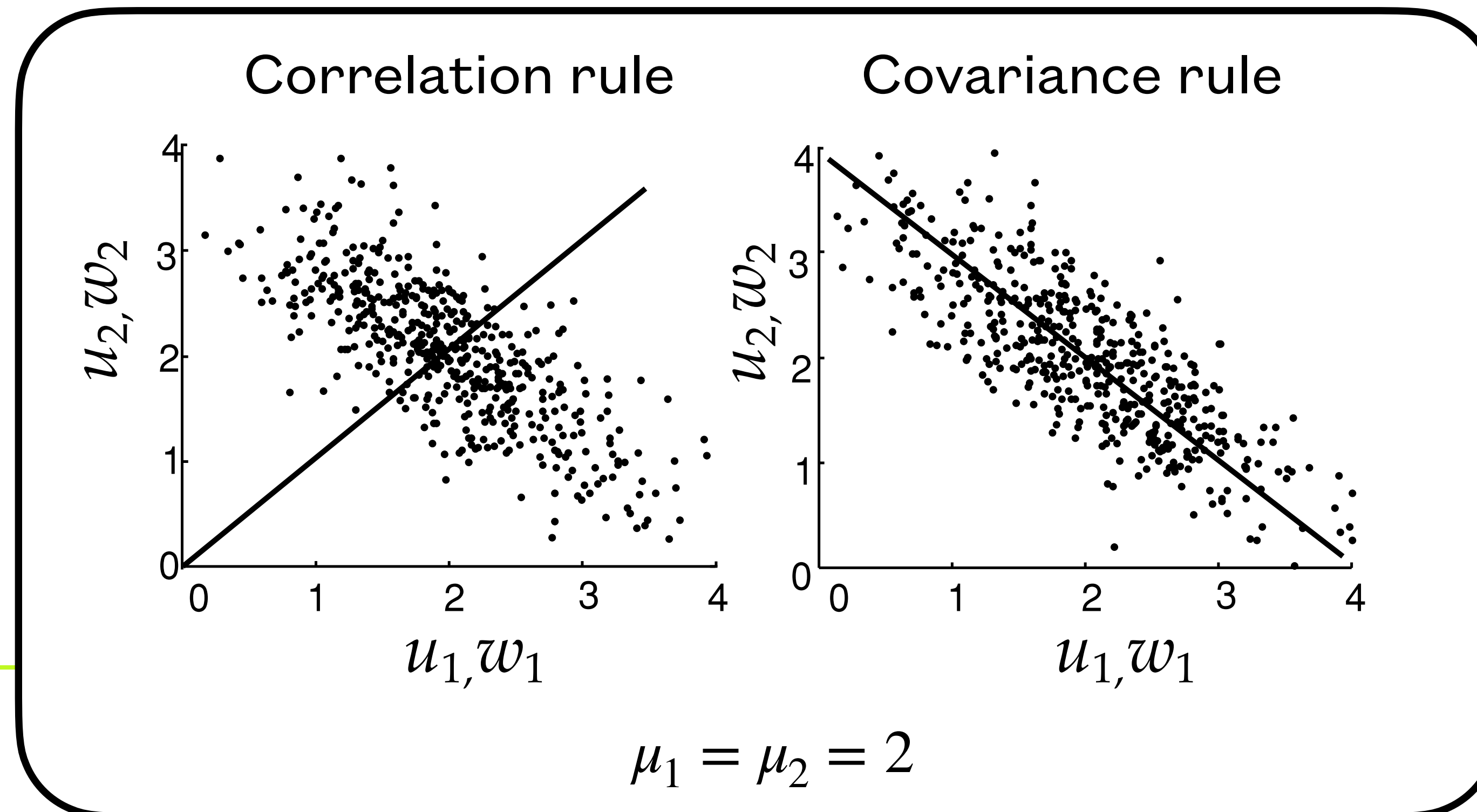
Learning rule: $W_{ij} = \sum_{\mu} \xi_i^\mu \xi_j^\mu$

Similarity between each pattern
and the current state

Hopfield network does PCA

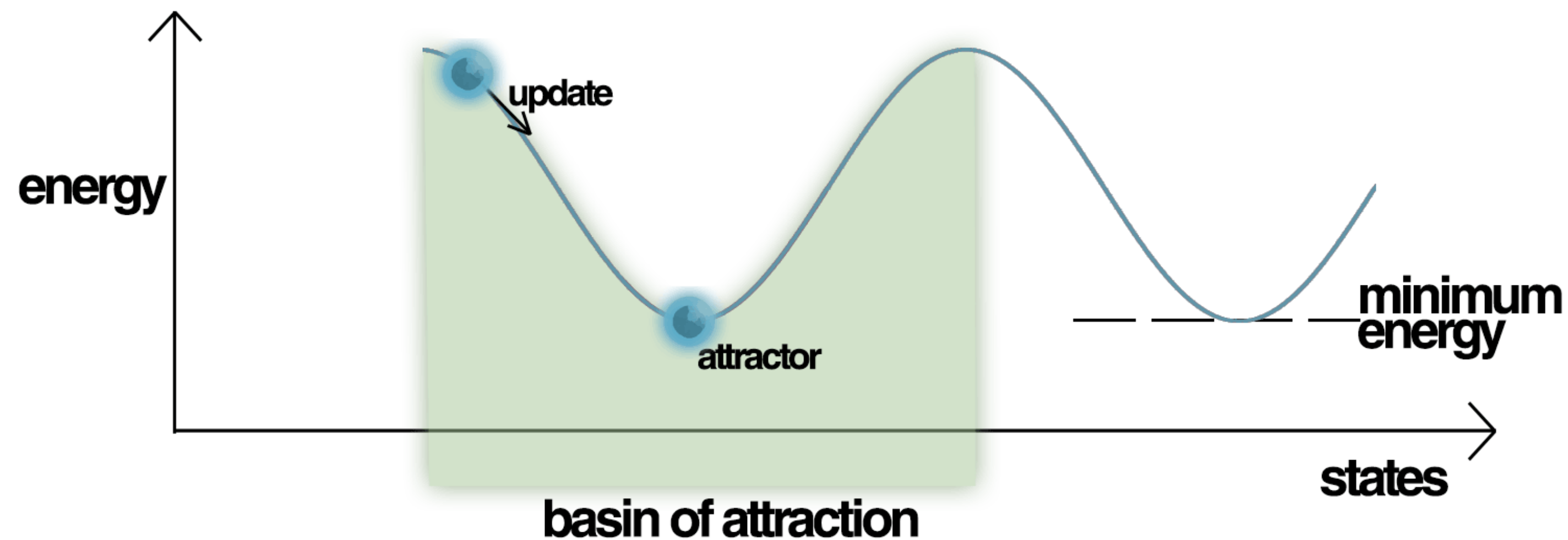
Consider each stored pattern ξ_i , assuming patterns are orthogonal to each other:

$$W_{\xi_i} = \frac{1}{N} \sum_{j=1}^P (\xi_j \xi_j^T) \xi_i = \frac{1}{N} (\xi_i \xi_i^T) \xi_i = \frac{1}{N} \|\xi_i\|^2 \xi_i = \frac{1}{N} (N \xi_i) = \xi_i$$



Hopfield network minimizes energy

Energy function (Lyapunov function) $E(s) = - \sum_{i,j} W_{ij} s_i s_j - \sum_i \theta_i s_i$



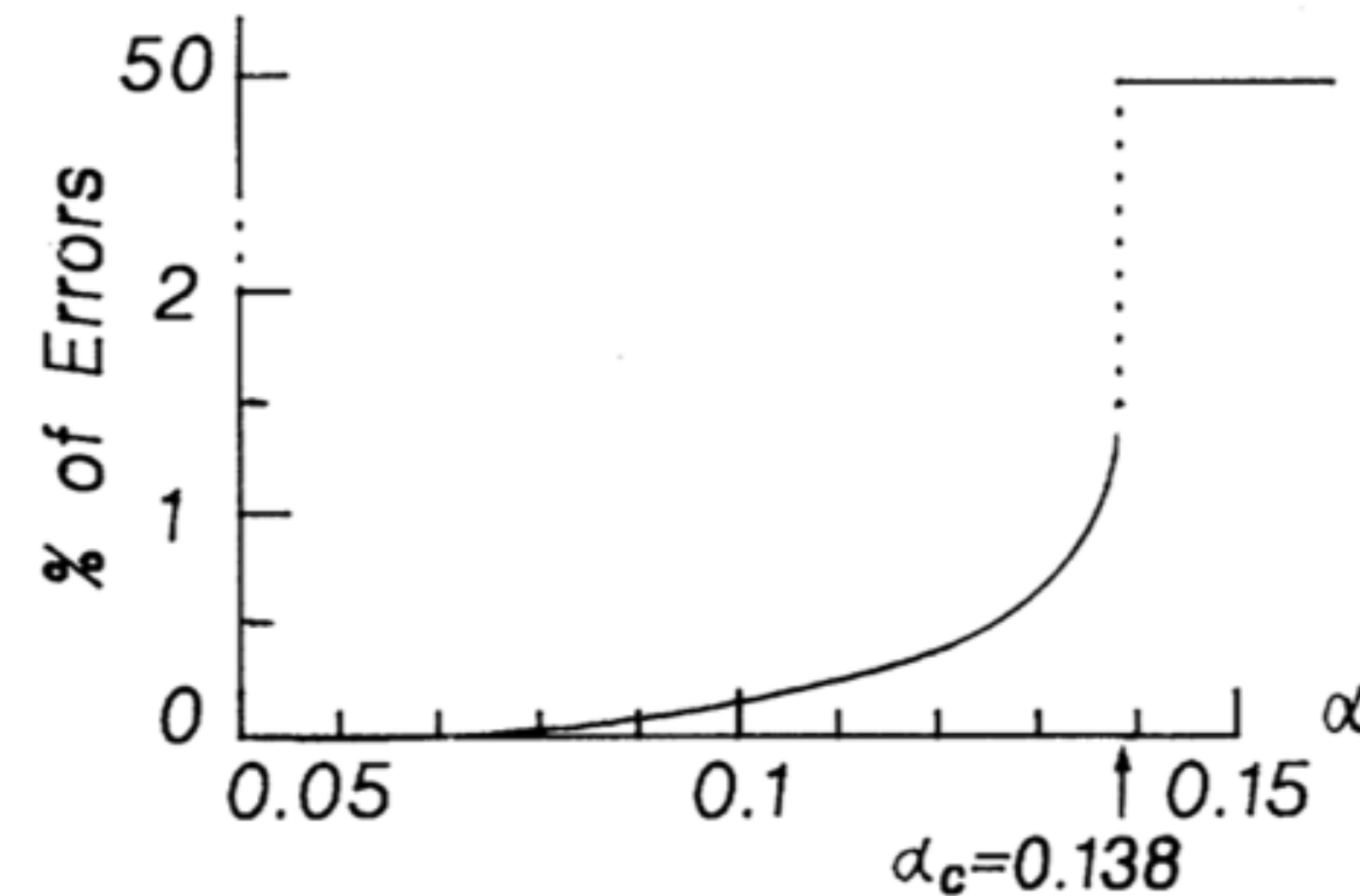
Hopfield network

N neurons

$\sim N^2$ synapses

$\sim 2^N$ possible patterns

How many can be stored?

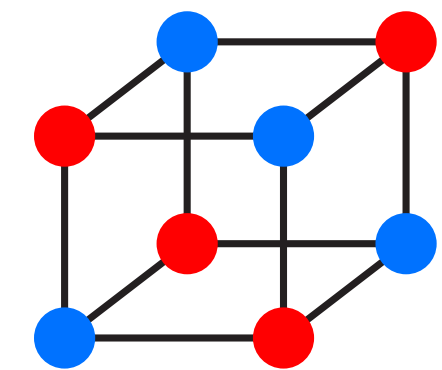


Error-free at
 $\sim 0.05 N$

Capacity
 $\sim 0.138 N$

Hopfield network vs. Ising model

Hopfield network	Ising Model (magnetism)
Neuron	Atom
Activation state	Spin (magnetic moment)
Threshold	Strength of outer magnetic field
Connection weights	Magnetic coupling



❖ Summary

- Synaptic mechanisms
 - LTP/LDP
 - STDP
- Hebbian Learning
 - Correlation/Covariance rule
 - Ocular dominance
 - Supervised Hebbian learning
- Hopfield Network

